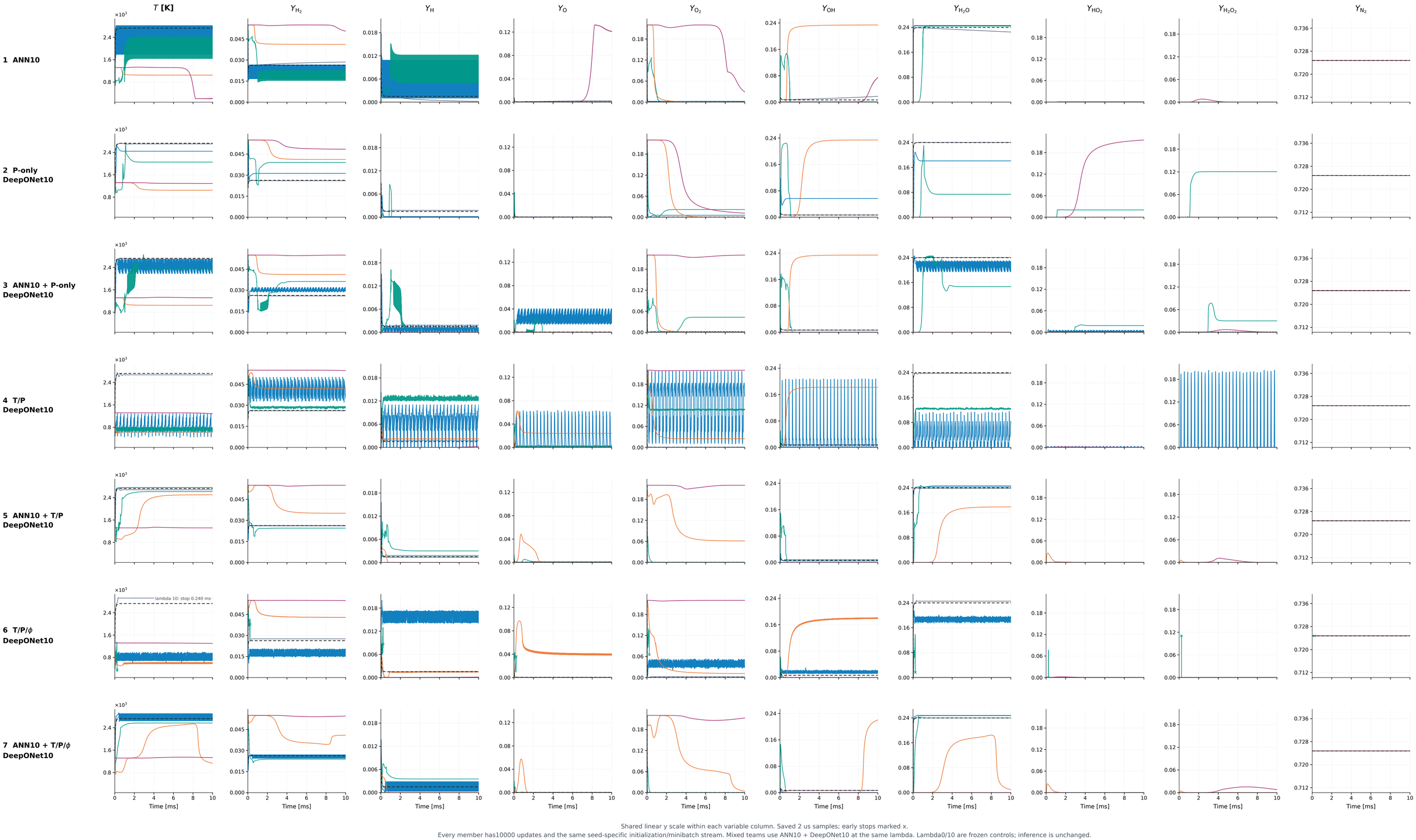
Element soft-constraint weight sweep | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 04 | Cantera vs NN NRMSE

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Repaired self-fed, 10 ms

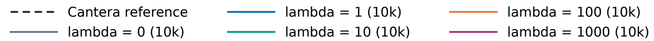
$$\text{NRMSE} = \sqrt{\text{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\text{mean}[y_{Cantera}^2]}$$

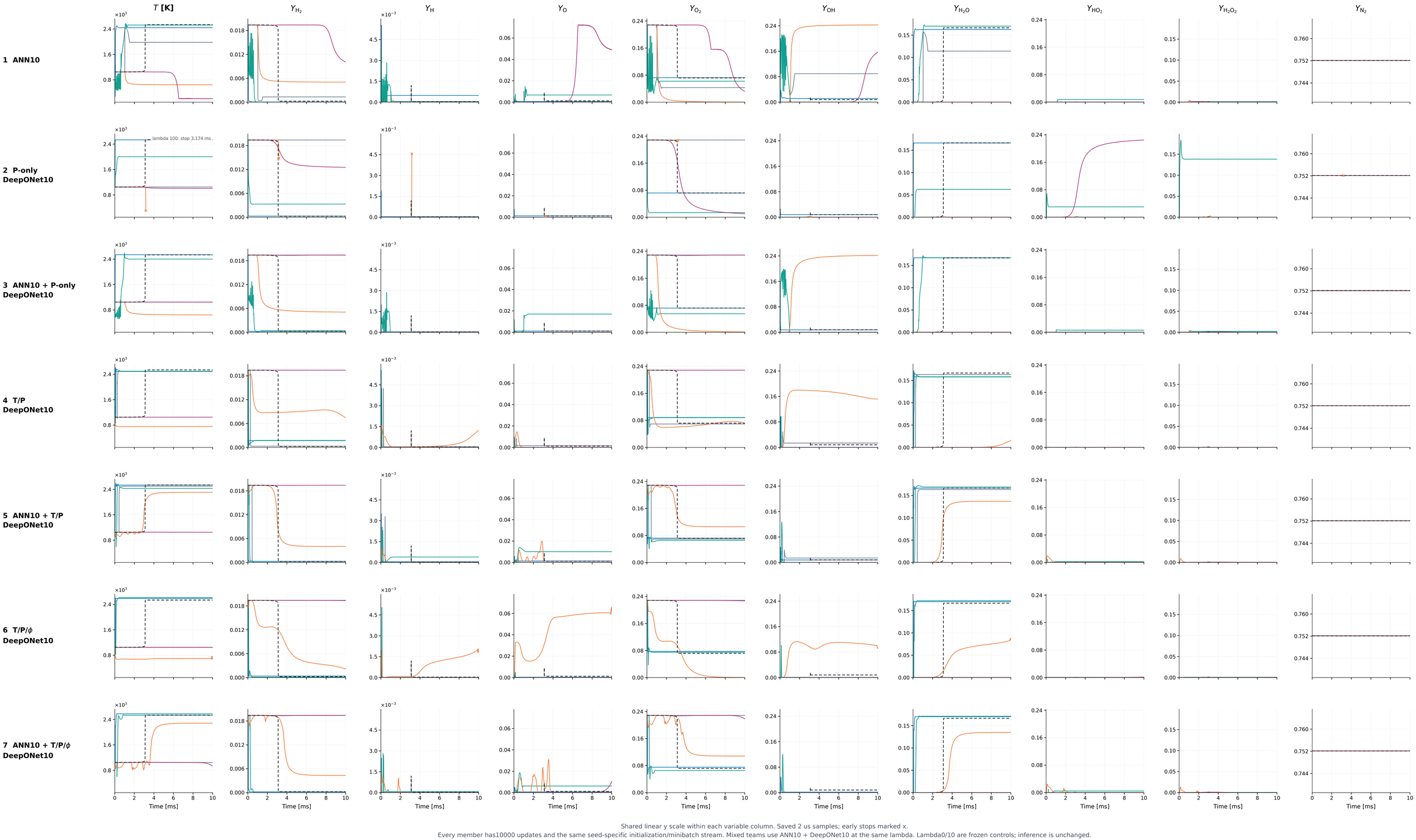
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0108	0.0583	0.593	2.33	0.0953	0.832	0.0354	0.149	0.418	0	Complete
1 ANN10	**1**	0.235	0.253	4.12	0.891	0.427	0.971	0.0296	28.3	551	0	Complete
1 ANN10	**10**	0.346	0.354	4.88	0.747	2.76	5.37	0.288	170	740	0	Complete
1 ANN10	**100**	0.608	0.621	1	1	5.54	31.4	1	1.07	109	0	Complete
1 ANN10	**1000**	0.623	1.07	1	100	18.8	2.47	1	0.998	2.38e+04	0	Complete
2 P-only DeepONet10	**0**	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	**1**	0.109	0.2	0.888	0.462	1.24	7.23	0.253	0.999	5.42	0	Complete
2 P-only DeepONet10	**10**	0.29	0.486	1.13	3.72	2.47	8.33	0.704	5.9e+03	8.86e+05	0	Complete
2 P-only DeepONet10	**100**	0.593	0.718	1	1	9.71	28.3	1	1.02	3.14	0	Complete
2 P-only DeepONet10	**1000**	0.522	0.942	1	1	12.3	1	1	4.93e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.12	0.151	0.679	45.4	0.614	0.988	0.118	933	3.44e+03	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.262	0.385	2.17	8.88	4.02	6.24	0.44	4.9e+03	2.39e+05	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.606	0.637	1	1	6.4	31	1	1.07	34	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.514	1.08	1	1	21	0.999	1	0.998	2.71e+04	0	Complete
4 T/P DeepONet10	**0**	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	**1**	0.703	0.597	3.61	40.5	13.7	7.74	0.844	278	5.21e+05	0	Complete
4 T/P DeepONet10	**10**	0.727	0.125	7.04	6.23	10.4	1.04	0.489	5.78	0.913	0	Complete
4 T/P DeepONet10	**100**	0.767	0.628	0.526	46.2	4.7	24.4	1	10.1	7.44	0	Complete
4 T/P DeepONet10	**1000**	0.517	1.09	0.982	1	21.2	1	1	268	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.0233	0.0207	0.323	0.481	0.182	0.356	0.0142	0.347	2	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.137	0.11	1.45	2.28	0.847	3.46	0.134	59.7	588	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.318	0.618	0.973	23.7	10.9	1	0.56	1.53e+03	6.86e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.514	1.08	1	0.998	21	0.999	1	2.72	3.47e+04	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	**1**	0.716	0.372	9.46	0.718	3.56	1.49	0.213	0.563	0.733	0	Complete
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.240 ms
6 T/P/ϕ DeepONet10	**100**	0.779	0.691	0.307	80	4.51	23	1	12	5.86	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.517	1.09	0.979	1	21.2	1	1	242	148	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.0447	0.0412	0.628	0.907	0.141	0.98	0.034	5.91	562	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.108	0.127	1.5	1.53	0.832	2.62	0.115	83.1	639	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.384	0.675	0.99	20.7	11.9	10.7	0.682	1.36e+03	5.05e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.51	1.07	1	1	20.6	1	1	1.78	7.63e+04	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 05 | Cantera vs NN NRMSE

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

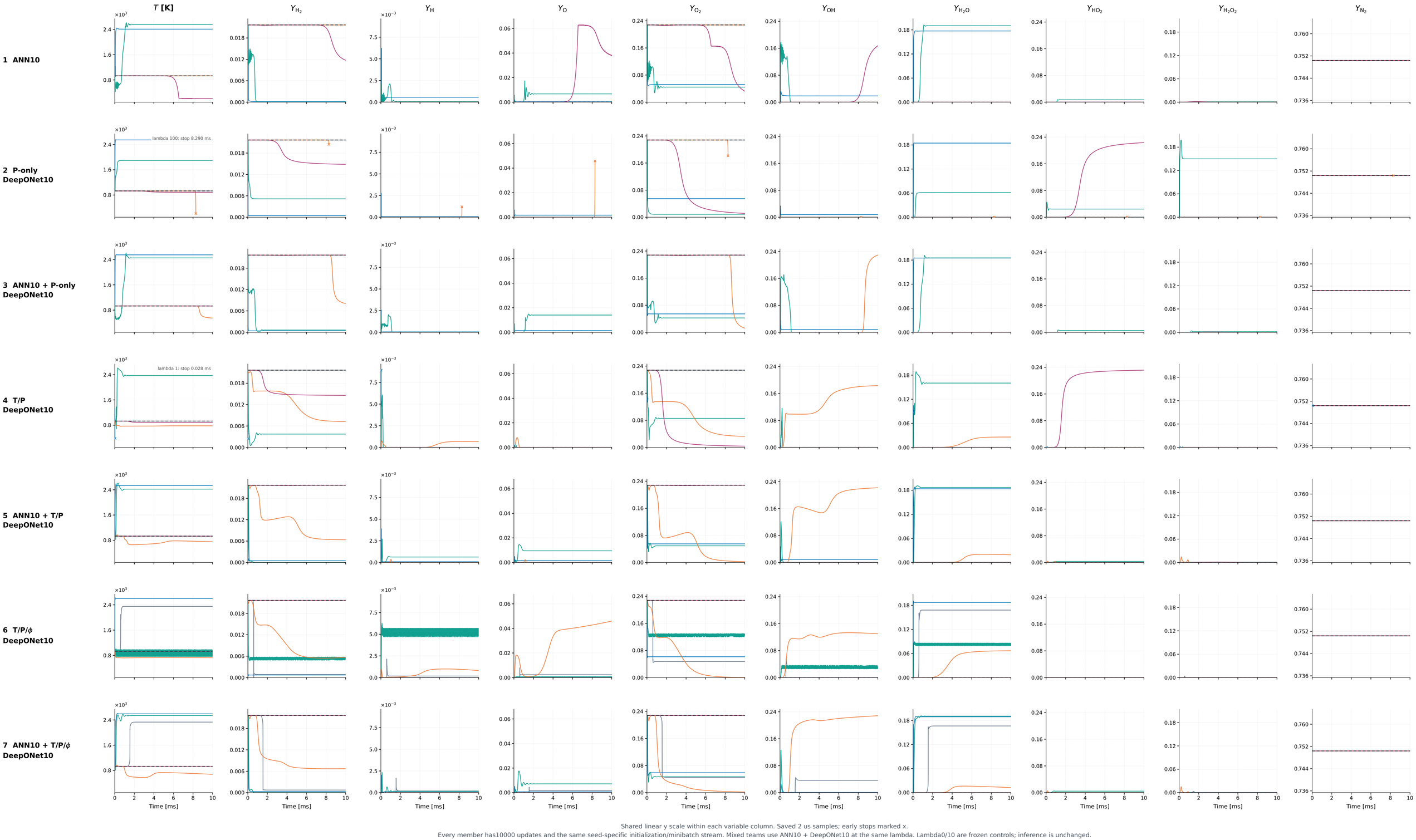
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.299	0.773	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
1 ANN10	**1**	0.352	0.987	14.1	0.633	0.613	1.11	0.646	2.54	0.981	0	Complete
1 ANN10	**10**	0.318	0.922	10.6	5.47	0.659	6.66	0.585	121	6.62	0	Complete
1 ANN10	**100**	0.722	0.669	1	1	0.798	32.4	1	0.961	3.53	0	Complete
1 ANN10	**1000**	0.752	1.36	1	37.2	0.686	6.97	1	1	2.97	0	Complete
2 P-only DeepONet10	**0**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	**1**	0.375	0.981	1.42	0.759	0.617	0.677	0.664	0.948	0.998	0	Complete
2 P-only DeepONet10	**10**	0.31	0.844	1	1	0.909	1	0.669	507	872	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 3.174 ms
2 P-only DeepONet10	**1000**	0.58	1	1	1	0.304	1	1	2.83e+03	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.376	0.982	1.51	0.685	0.617	0.665	0.666	0.903	0.997	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.301	0.912	10.3	15	0.687	6.74	0.576	96.3	13.6	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.72	0.641	1	1	0.746	31.6	1	0.954	1.57	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.564	1.47	1	0.996	0.926	1	1	0.998	3.21	0	Complete
4 T/P DeepONet10	**0**	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	**1**	0.364	0.915	4.26	1.05	0.563	1.11	0.633	0.92	0.959	0	Complete
4 T/P DeepONet10	**10**	0.36	0.909	7.05	1.02	0.56	1.25	0.627	1.03	1	0	Complete
4 T/P DeepONet10	**100**	0.682	0.831	12.4	2.5	0.593	23.5	0.987	1	1	0	Complete
4 T/P DeepONet10	**1000**	0.565	1.48	1	1	0.928	1	1	1.48	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.374	0.979	2.56	0.739	0.615	0.71	0.663	0.884	0.997	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.342	0.968	11.8	9.17	0.631	1.8	0.656	42.5	1.31	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.106	0.336	3.97	3.99	0.236	1	0.21	55.1	7.6	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.565	1.47	1	0.993	0.927	1	1	0.997	2.21	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**1**	0.388	0.971	1	0.782	0.593	1	0.676	0.995	1	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.388	0.976	8.65	1.03	0.598	1.29	0.675	3.94	3.85	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.706	0.481	31.8	47.7	0.553	14	0.608	1	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.565	1.47	1	1	0.927	1	1	5.67	1.27	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.382	0.978	2.2	0.683	0.603	0.962	0.674	0.89	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.354	0.952	6.19	5.75	0.621	1.69	0.651	74.1	3.15	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.2	0.523	5.71	6.17	0.331	1	0.342	68.2	9.82	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.567	1.47	1	1.53	0.923	1	1	0.995	2.73	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 07 | Cantera vs NN NRMSE

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

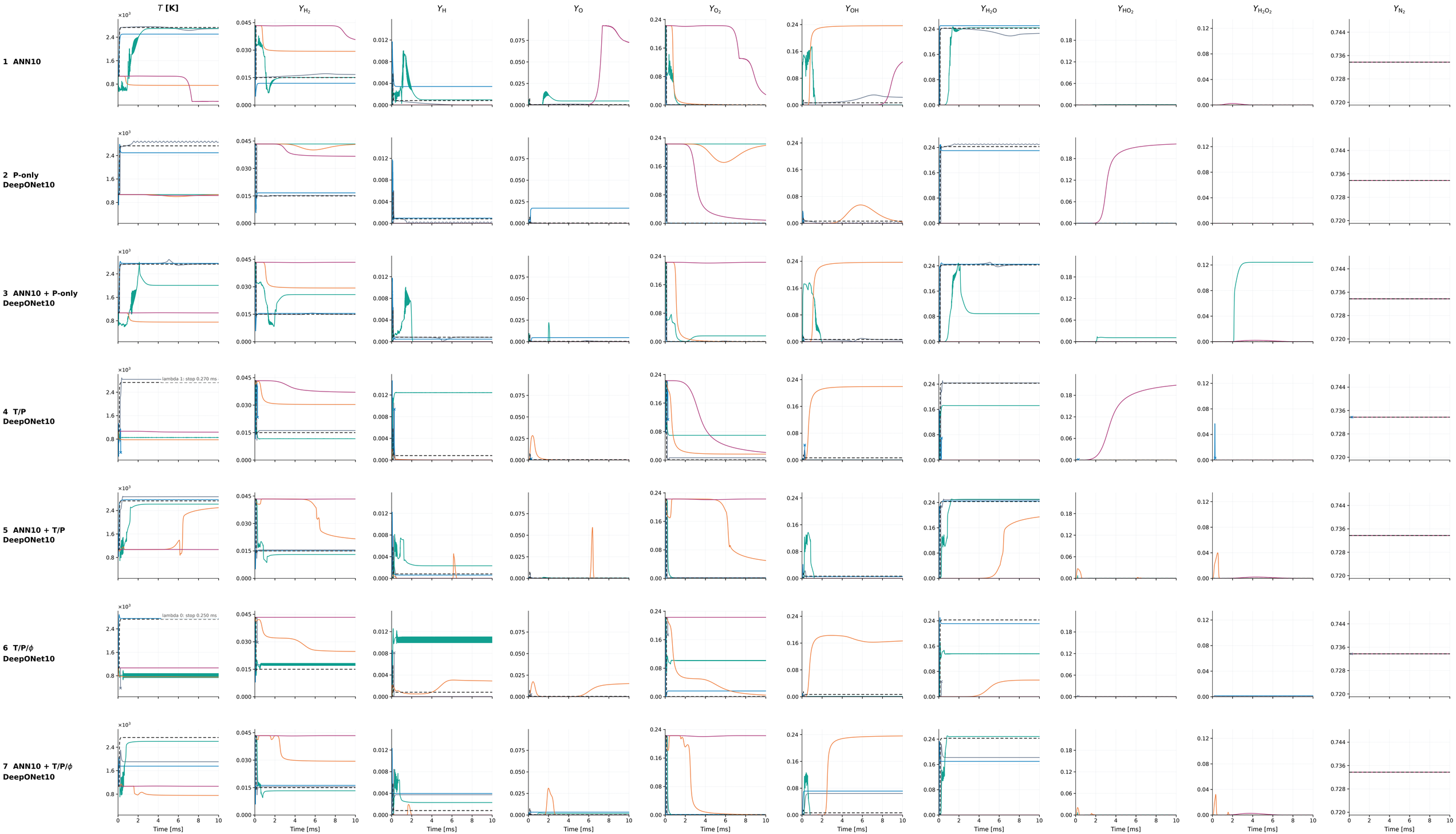
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
1 ANN10	**1**	1.58	0.99	1.60e+08	9.77e+07	0.769	4.13e+09	1.06e+06	199	18.3	0	Complete
1 ANN10	**10**	1.66	0.968	1.05e+08	8.70e+08	0.791	8.82e+09	1.09e+06	1.38e+04	2.24e+03	0	Complete
1 ANN10	**100**	2.84e-06	2.61e-06	1	51.2	4.32e-06	1	1	0.999	1	0	Complete
1 ANN10	**1000**	0.463	0.147	1	4.38e+09	0.323	1.19e+10	1	1	1.35e+03	0	Complete
2 P-only DeepONet10	**0**	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	**1**	1.73	0.977	2.26e+07	2.25e+08	0.758	1.77e+09	1.11e+06	52.6	7.03	0	Complete
2 P-only DeepONet10	**10**	1.03	0.756	1.40e+05	1.48e+05	0.958	1.20e+05	3.63e+05	5.08e+04	3.81e+05	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 8.290 ms
2 P-only DeepONet10	**1000**	0.0356	0.235	1	1	0.71	1	1	3.45e+05	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	**1**	1.74	0.979	2.24e+07	1.91e+08	0.76	1.88e+09	1.11e+06	54.5	5.62	0	Complete
3 ANN10 + P-only DeepONet10	**10**	1.56	0.942	1.11e+08	1.77e+09	0.802	9.78e+09	1.06e+06	1.04e+04	4.12e+03	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.136	0.209	26.1	409	0.315	1.69e+10	1	39.1	272	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	1.54e-03	1.32e-03	1	3.98e+06	1.99e-03	2.07e+04	1	0.999	1.22e+03	0	Complete
4 T/P DeepONet10	**0**	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
4 T/P DeepONet10	**10**	1.54	0.832	1.76e+08	1.19e+07	0.628	2.08e+09	9.60e+05	70.1	285	0	Complete
4 T/P DeepONet10	**100**	0.164	0.487	1.21e+08	1.62e+08	0.65	3.18e+10	9.96e+04	1	1	0	Complete
4 T/P DeepONet10	**1000**	0.0435	0.287	1	1	0.867	1	1	4.21e+05	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	**1**	1.72	0.975	3.66e+07	2.04e+08	0.757	2.00e+09	1.10e+06	37.9	2.61	0	Complete
5 ANN10 + T/P DeepONet10	**10**	1.59	0.993	1.77e+08	1.28e+09	0.781	2.22e+09	1.11e+06	6.2e+03	663	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.207	0.548	1.07e+07	3.08e+07	0.78	3.95e+10	8.07e+04	1.02e+03	4.11e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	1.55e-03	1.36e-03	1	1.32e+06	2.05e-03	1.48e+04	1	0.996	1.26e+03	0	Complete
6 T/P/ϕ DeepONet10	**0**	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	**1**	1.79	0.966	2.92e+06	3.23e+07	0.73	5.19e+05	1.12e+06	11.5	1.01	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.0976	0.755	1.48e+09	6.75e+07	0.453	6.92e+09	4.97e+05	2.18	1	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.218	0.606	1.99e+08	4.47e+09	0.819	2.70e+10	3.01e+05	1	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	6.13e-06	2.48e-05	1	0.974	7.69e-05	1	1	38.1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	1.38	0.888	5.23e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	1.78	0.98	3.71e+07	1.04e+08	0.74	5.22e+08	1.13e+06	30.4	1.59	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	1.71	0.993	6.20e+07	9.94e+08	0.791	2.27e+09	1.13e+06	8.85e+03	1.14e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.264	0.613	4.48e+05	3.08e+06	0.885	4.54e+10	6.99e+04	1.04e+03	3.59e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	6.38e-04	5.63e-04	1	1	8.49e-04	1	1	0.603	521	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 18 | Cantera vs NN NRMSE

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Repaired self-fed, 10 ms

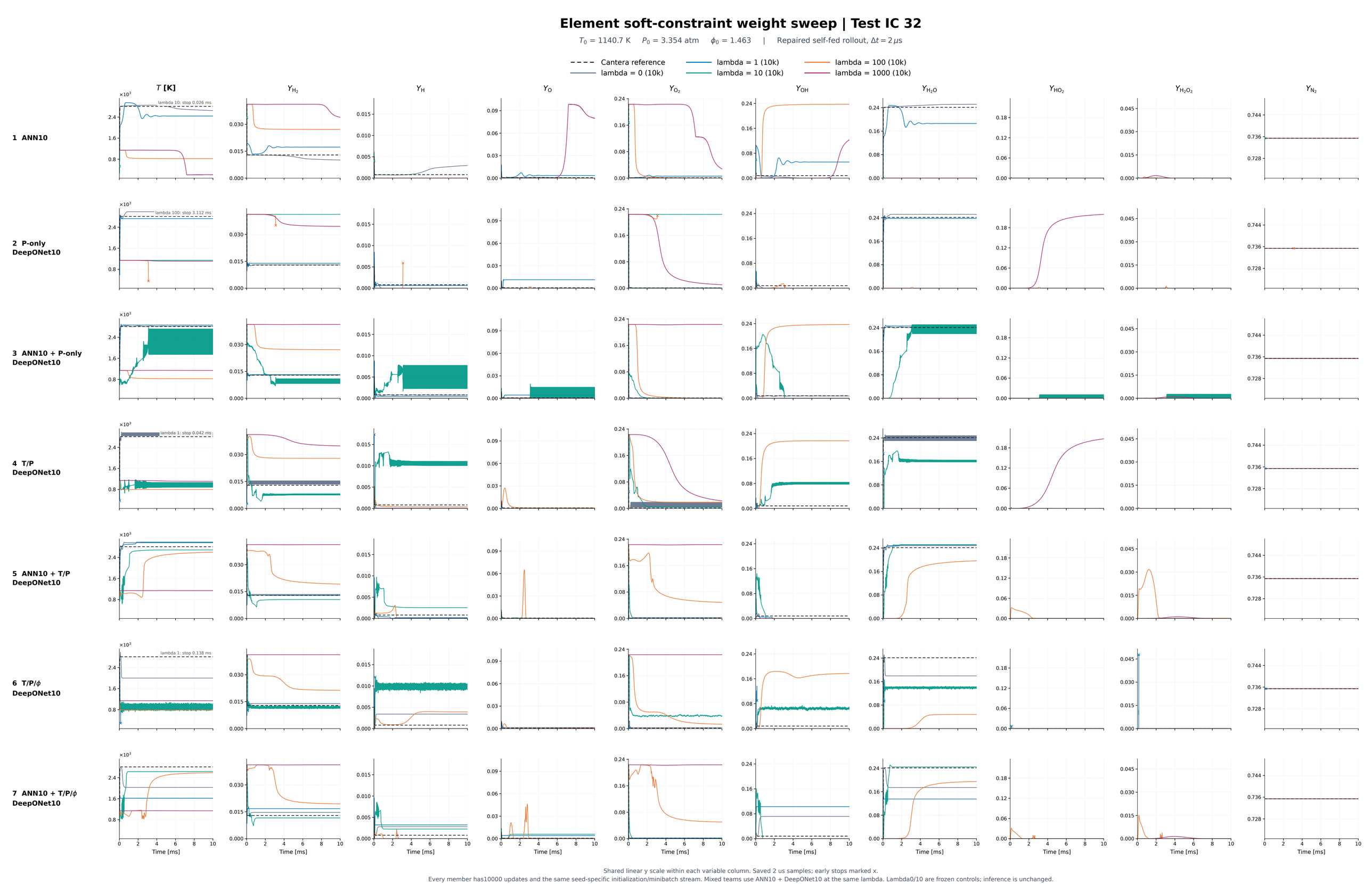
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
1 ANN10	**1**	0.0856	0.274	3.21	0.99	0.697	1	0.0878	2.71	73.7	0	Complete
1 ANN10	**10**	0.249	0.34	2.24	10	1.06	6.81	0.295	75.5	106	0	Complete
1 ANN10	**100**	0.714	1	1	1	2.24	32.8	1	1.04	58.6	0	Complete
1 ANN10	**1000**	0.714	1.75	1	89.1	7.69	5.29	1	1	756	0	Complete
2 P-only DeepONet10	**0**	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	**1**	0.0869	0.199	0.927	34.1	0.631	1.02	0.0847	34.5	2.07	0	Complete
2 P-only DeepONet10	**10**	0.608	1.8	1	1	8.84	1	1	1	1	0	Complete
2 P-only DeepONet10	**100**	0.617	1.73	1	1	8.16	3.44	1	1.05	1.07	0	Complete
2 P-only DeepONet10	**1000**	0.616	1.52	1	1	4.73	1	1	1.21e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0228	0.178	0.954	8.55	0.672	0.981	0.0743	13.8	1.83	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.351	0.681	2.4	4.07	0.985	8.15	0.637	706	9.43e+04	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.711	1.04	1	1	2.72	32.1	1	1.03	15.8	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.607	1.8	1	1	8.81	1	1	1	1.01e+03	0	Complete
4 T/P DeepONet10	**0**	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.270 ms
4 T/P DeepONet10	**10**	0.687	0.221	13.7	0.996	2.76	0.914	0.293	2.67	0.999	0	Complete
4 T/P DeepONet10	**100**	0.714	1.04	0.96	10.9	1.91	30.3	1	1.02	1	0	Complete
4 T/P DeepONet10	**1000**	0.615	1.56	1	1	4.96	1	1	1.07e+04	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.0329	0.184	0.893	0.514	0.761	0.408	0.0793	1.11	0.974	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.172	0.24	2.57	0.648	0.895	4.41	0.175	24.9	70.6	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.484	1.42	1.06	13.3	6.91	1	0.795	331	5.83e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.607	1.8	1	1	8.81	1	1	1	913	0	Complete
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.250 ms
6 T/P/ϕ DeepONet10	**1**	0.0553	0.226	1.04	1.01	0.995	1	0.102	23	1.28e+03	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.703	0.185	11.5	0.954	4.01	0.874	0.441	2.19	1	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.72	0.936	1.83	17.6	2.47	23.7	0.897	1.03	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.608	1.8	1	1	8.84	1	1	1.17	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.358	0.195	3.78	5.81	0.784	9.69	0.311	1.18	1.15	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.143	0.214	2.3	2.77	0.856	3.11	0.142	28.9	47.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.702	1.19	0.99	12.7	4.1	29	1	187	3.31e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.607	1.8	1	0.991	8.81	1	1	1	1.06e+03	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.

Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 32 | Cantera vs NN NRMSE

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Repaired self-fed, 10 ms

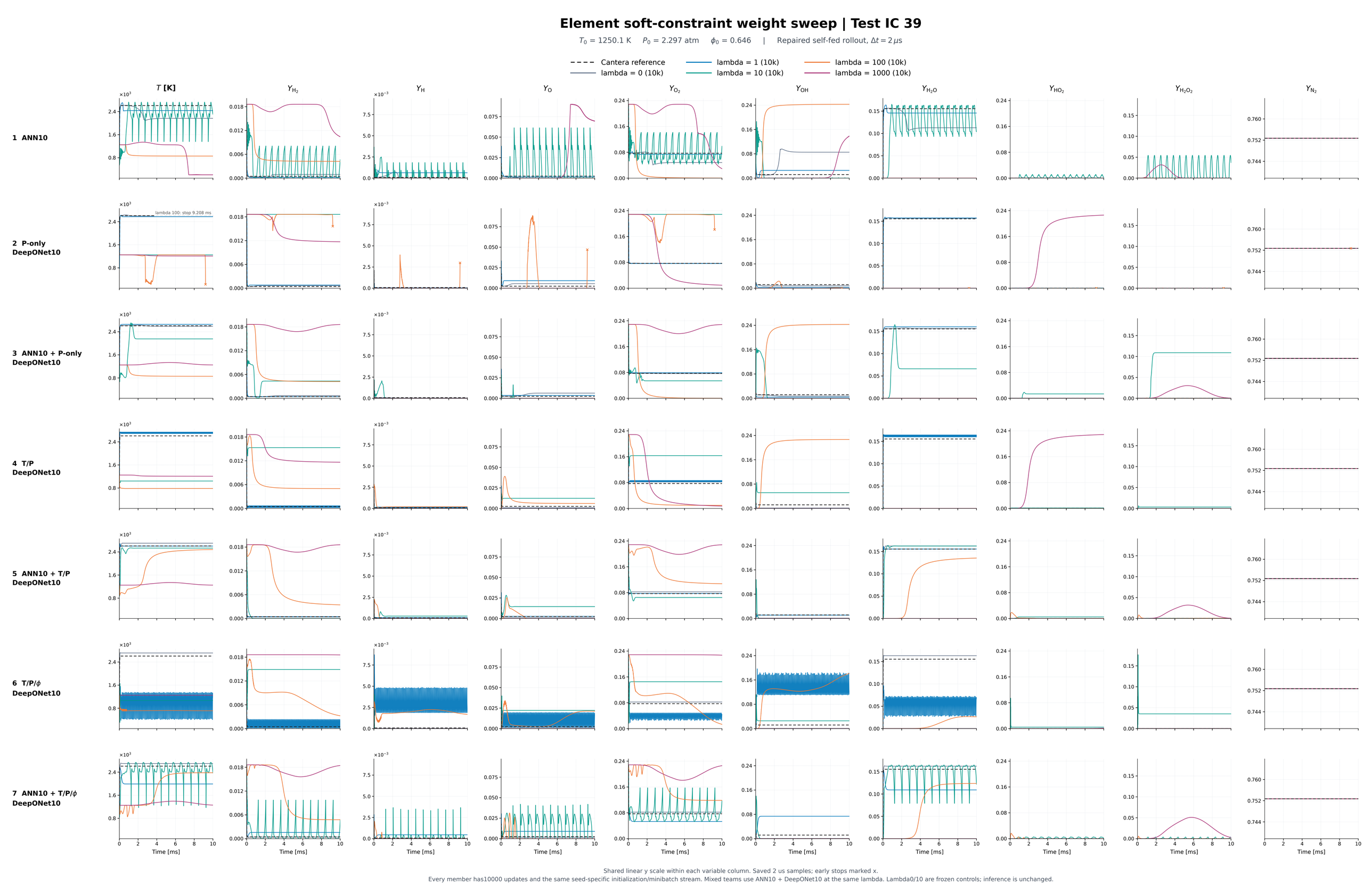
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
1 ANN10	**1**	0.126	0.312	0.991	4.6	0.374	5.21	0.216	0.765	0.904	0	Complete
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
1 ANN10	**100**	0.698	1.19	1	1	4.15	26.9	1	1.26	156	0	Complete
1 ANN10	**1000**	0.712	2.09	1	78.4	13.9	4.19	1	0.999	653	0	Complete
2 P-only DeepONet10	**0**	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	**1**	0.0366	0.0924	0.352	16.8	0.392	0.997	0.0296	10.6	2.02	0	Complete
2 P-only DeepONet10	**10**	0.594	2.15	1	1	16.2	1	1	1.05	2.87	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 3.112 ms
2 P-only DeepONet10	**1000**	0.601	1.84	1	1	9.17	1	1	1.49e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0308	0.0574	0.59	5.83	0.385	1.03	0.033	3.59	3.04	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.395	0.466	5.64	13.2	1.17	9.18	0.386	543	1.75e+03	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.694	1.24	1	1	4.87	26.3	1	1.17	43.5	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.593	2.15	1	0.997	16.1	1	1	0.999	608	0	Complete
4 T/P DeepONet10	**0**	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.042 ms
4 T/P DeepONet10	**10**	0.657	0.458	12.2	0.996	2.43	8.18	0.341	24.4	1.53	0	Complete
4 T/P DeepONet10	**100**	0.716	1.22	0.553	7.62	3.7	24.6	1	0.599	0.893	0	Complete
4 T/P DeepONet10	**1000**	0.6	1.91	1	1	10.3	1	1	1.23e+04	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.06	0.0683	0.971	0.981	0.395	1.01	0.0468	0.973	3.89	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.162	0.248	3.14	0.602	0.721	3.12	0.142	31.1	91	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.338	1.07	1.02	13.4	7.48	1	0.534	910	1.28e+04	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.593	2.15	1	1	16.1	1	1	0.999	609	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.138 ms
6 T/P/ϕ DeepONet10	**10**	0.679	0.207	11.1	0.879	3.13	6.86	0.428	5.83	1.02	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.721	1.01	2.96	1.62	4.44	19.8	0.883	0.381	0.845	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.594	2.15	1	1	16.2	1	1	1.47	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.424	0.3	2.99	5.91	0.484	11.7	0.44	1.73	2.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.154	0.173	2.44	8.16	0.604	3.25	0.134	29.5	33.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.353	1.24	0.948	11.3	8.95	1	0.578	540	2.88e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.593	2.15	1	0.998	16.1	1	1	0.999	752	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.

Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 39 | Cantera vs NN NRMSE

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Repaired self-fed, 10 ms

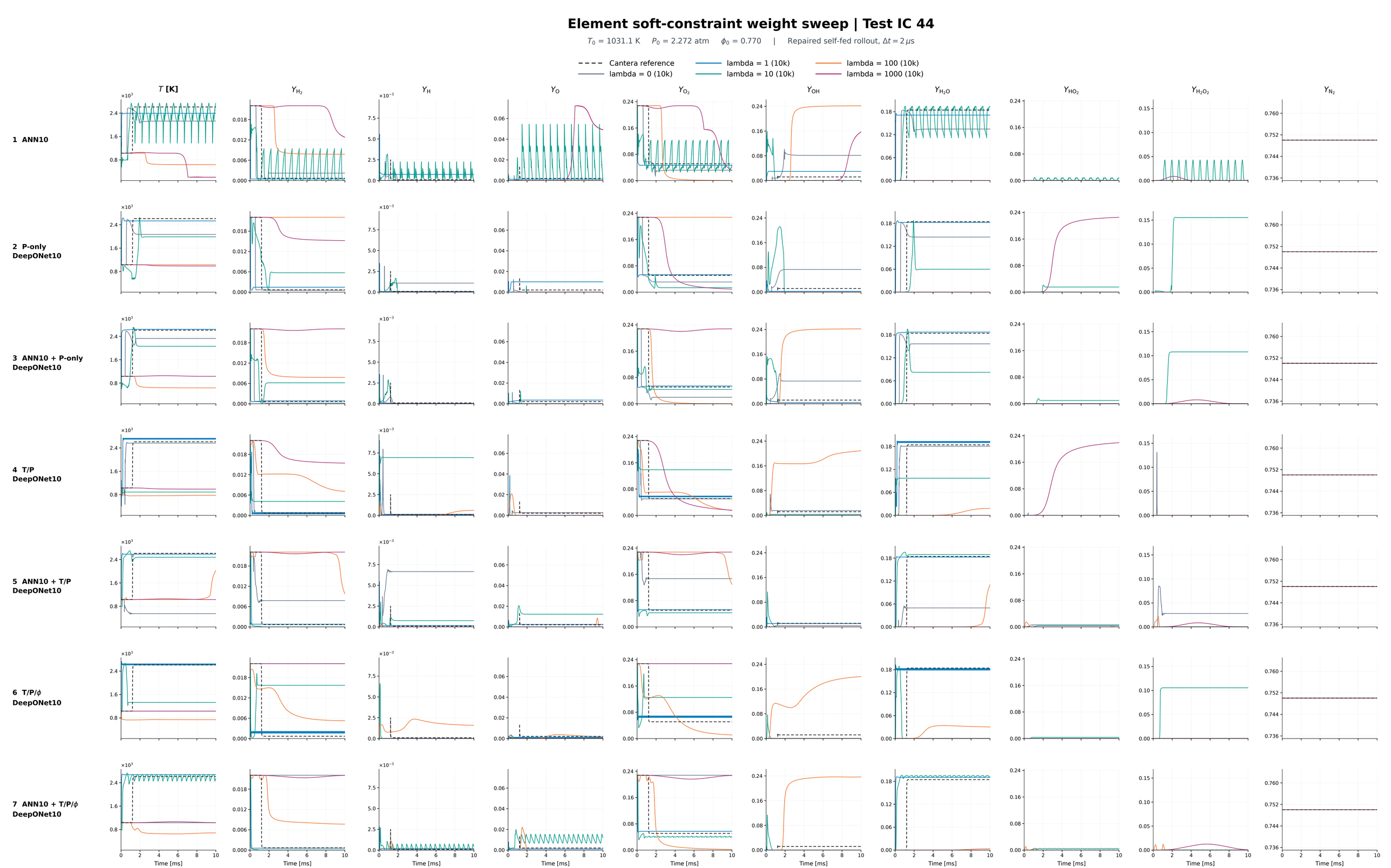
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
1 ANN10	**1**	0.0665	0.517	5.92	0.548	0.0509	1.15	0.0653	2.65	2.51	0	Complete
1 ANN10	**10**	0.216	4.61	6	8.07	0.391	2.82	0.329	286	2.53e+04	0	Complete
1 ANN10	**100**	0.658	7.05	1	1	1.05	18.6	1	0.909	214	0	Complete
1 ANN10	**1000**	0.651	19.1	1	16.3	1.59	3.32	1	0.991	1.26e+04	0	Complete
2 P-only DeepONet10	**0**	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	**1**	0.0247	0.904	0.643	2.66	0.0687	0.732	0.0409	15	3.55	0	Complete
2 P-only DeepONet10	**10**	0.52	20.5	1	1	1.95	1	1	0.908	0.791	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 9.208 ms
2 P-only DeepONet10	**1000**	0.531	15.7	1	1	1.18	1	1	9.64e+03	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0237	0.497	0.701	0.516	0.0415	0.668	0.0389	1.98	0.469	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.252	4.86	4.23	1.06	0.29	3.59	0.601	700	1.01e+05	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.654	7.79	1	1	1.07	18.2	1	1.05	103	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.506	19.7	1	1	1.8	0.999	1	0.979	1.72e+04	0	Complete
4 T/P DeepONet10	**0**	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	**1**	0.0408	0.201	0.703	0.966	0.0937	0.998	0.0467	4.83	7.85	0	Complete
4 T/P DeepONet10	**10**	0.6	16.8	0.946	3.92	1.11	3.4	0.999	72.2	3.24e+03	0	Complete
4 T/P DeepONet10	**100**	0.699	7.11	3.38	3.36	0.876	17.2	1	1.44	1.16	0	Complete
4 T/P DeepONet10	**1000**	0.532	14.7	1	1	1.05	1	1	1.05e+04	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	**1**	6.23e-03	0.116	0.353	0.176	9.54e-03	0.0848	7.54e-03	0.113	0.07	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.0613	1.32	2.39	4.95	0.172	1.21	0.0981	253	372	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.303	11.1	3.75	1.97	1.02	1	0.541	191	1.18e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.505	19.6	1	1	1.79	0.999	1	0.976	1.82e+04	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	**1**	0.624	1.27	32.7	4.23	0.499	10.7	0.638	4.51	8.29	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.591	16.2	1.53	7.82	0.873	1.23	1	299	3.62e+04	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.724	9.06	20.3	4.6	0.563	10.4	0.936	1.16	1.02	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.52	20.5	1	1	1.95	1	1	18.9	7.53	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.234	1.25	4.06	2.53	0.303	5.21	0.294	0.603	0.124	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.116	2.76	4.81	6.51	0.303	1.37	0.156	163	1.43e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.345	13.2	2.4	2.25	1.25	1	0.643	134	619	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.495	19	1	1	1.68	1	1	0.949	3.01e+04	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.

Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 44 | Cantera vs NN NRMSE

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
1 ANN10	**1**	0.203	0.953	6.1	0.491	0.663	1.76	0.344	1.47	0.99	0	Complete
1 ANN10	**10**	0.191	0.812	6.86	8.51	0.613	2.9	0.287	163	2.13e+03	0	Complete
1 ANN10	**100**	0.724	1.32	1	1	0.832	17.5	1	0.984	26.5	0	Complete
1 ANN10	**1000**	0.725	2.45	1	17	1.47	4.14	1	0.999	398	0	Complete
2 P-only DeepONet10	**0**	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	**1**	0.21	0.92	1.05	4.02	0.643	0.804	0.359	8.17	1.03	0	Complete
2 P-only DeepONet10	**10**	0.291	0.676	2.81	1	0.519	5.31	0.681	504	1.70e+04	0	Complete
2 P-only DeepONet10	**100**	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
2 P-only DeepONet10	**1000**	0.611	1.99	1	1	0.806	1	1	6.37e+03	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.218	0.942	1.19	0.847	0.646	0.715	0.365	1.53	1	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.214	0.867	4.32	1.02	0.538	3.95	0.556	337	1.22e+04	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.74	1.01	1	1	0.565	18.7	1	0.963	7.61	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.595	2.58	1	1	1.76	1	1	0.997	529	0	Complete
4 T/P DeepONet10	**0**	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	**1**	0.221	0.902	3.95	1.37	0.596	1	0.359	2.18	1.9	0	Complete
4 T/P DeepONet10	**10**	0.652	0.881	62.2	0.999	0.943	0.744	0.509	0.93	1	0	Complete
4 T/P DeepONet10	**100**	0.698	1.24	2.51	2.13	0.466	15	0.968	1	1	0	Complete
4 T/P DeepONet10	**1000**	0.61	2.01	1	1	0.713	1	1	6.08e+03	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.214	0.943	1.58	0.53	0.648	0.383	0.361	0.844	0.978	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.203	0.927	6.14	5.06	0.655	1.4	0.347	139	39.9	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.582	2.53	1.09	1.04	1.74	1	0.972	88.6	407	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.595	2.57	1	1	1.76	1	1	0.996	557	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**1**	0.222	0.912	0.975	0.782	0.618	1	0.361	0.959	0.999	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.505	1.94	5.23	1.04	0.896	1.14	1.03	120	1.25e+04	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.711	1.03	14.6	0.705	0.539	13.1	0.863	1	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.599	2.6	1	1	1.79	1	1	0.813	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.227	0.951	1.28	0.518	0.632	0.982	0.377	0.841	0.992	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.199	0.937	3.34	4.49	0.672	1.43	0.353	161	45.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.732	1.13	1.55	1.97	0.595	17.9	0.995	51.2	239	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.591	2.55	1	1	1.73	1	1	0.994	908	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 45 | Cantera vs NN NRMSE

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
1 ANN10	**1**	0.128	0.799	1.01	1.56	0.664	2.46	0.211	1.01	0.647	0	Complete
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
1 ANN10	**100**	0.702	2.97	1	1	4.13	14.6	1	1.5	229	0	Complete
1 ANN10	**1000**	0.722	5.45	1	25.3	13.5	2.47	1	0.996	188	0	Complete
2 P-only DeepONet10	**0**	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	**1**	0.0717	0.228	1.31	8.8	0.519	0.872	0.0335	9.57	1.59	0	Complete
2 P-only DeepONet10	**10**	0.574	1.23	6.93	1.23	1.44	8.57	0.609	16.1	583	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 1.658 ms
2 P-only DeepONet10	**1000**	0.6	4.75	1	1	8.85	1	1	1.21e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0203	0.215	0.247	3.61	0.59	0.962	0.0664	0.914	1	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.311	1.05	2.13	15	0.696	4.61	0.35	399	5.68e+03	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.641	2.94	1	1	4.64	13	0.93	1.63	78.9	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.591	5.63	1	0.999	15.9	1	1	0.992	200	0	Complete
4 T/P DeepONet10	**0**	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	**1**	0.07	0.17	1.01	0.997	0.525	0.998	0.065	9.21	41.6	0	Complete
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
4 T/P DeepONet10	**100**	0.74	3.2	1.3	1.91	4.36	11.4	1	1.66	0.897	0	Complete
4 T/P DeepONet10	**1000**	0.592	5.63	1	1	16	1	1	0.803	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.1	0.393	0.791	1.02	0.723	1.04	0.121	0.91	2.62	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.684	4.53	0.583	12.1	7.96	2.26	1	626	2.30e+04	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.355	3.13	1.08	5.45	8.23	1	0.55	316	1.03e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.591	5.63	1	0.999	15.9	1	1	0.988	118	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.134 ms
6 T/P/ϕ DeepONet10	**10**	0.511	4.29	2.03	1.36	8.35	1.52	0.993	750	7.11e+04	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.755	3.2	1.08	25.7	3.81	7.77	0.856	1.98	0.906	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.592	5.63	1	1	16	1	1	0.897	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.448	1.19	2.59	3.51	0.432	7.24	0.522	1.67	1.27	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.0934	0.457	2.4	5.81	0.839	1.42	0.108	63.1	21	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.349	3.05	1.15	5.29	8.24	1	0.543	358	1.15e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.591	5.63	1	0.996	15.9	1	1	0.978	219	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

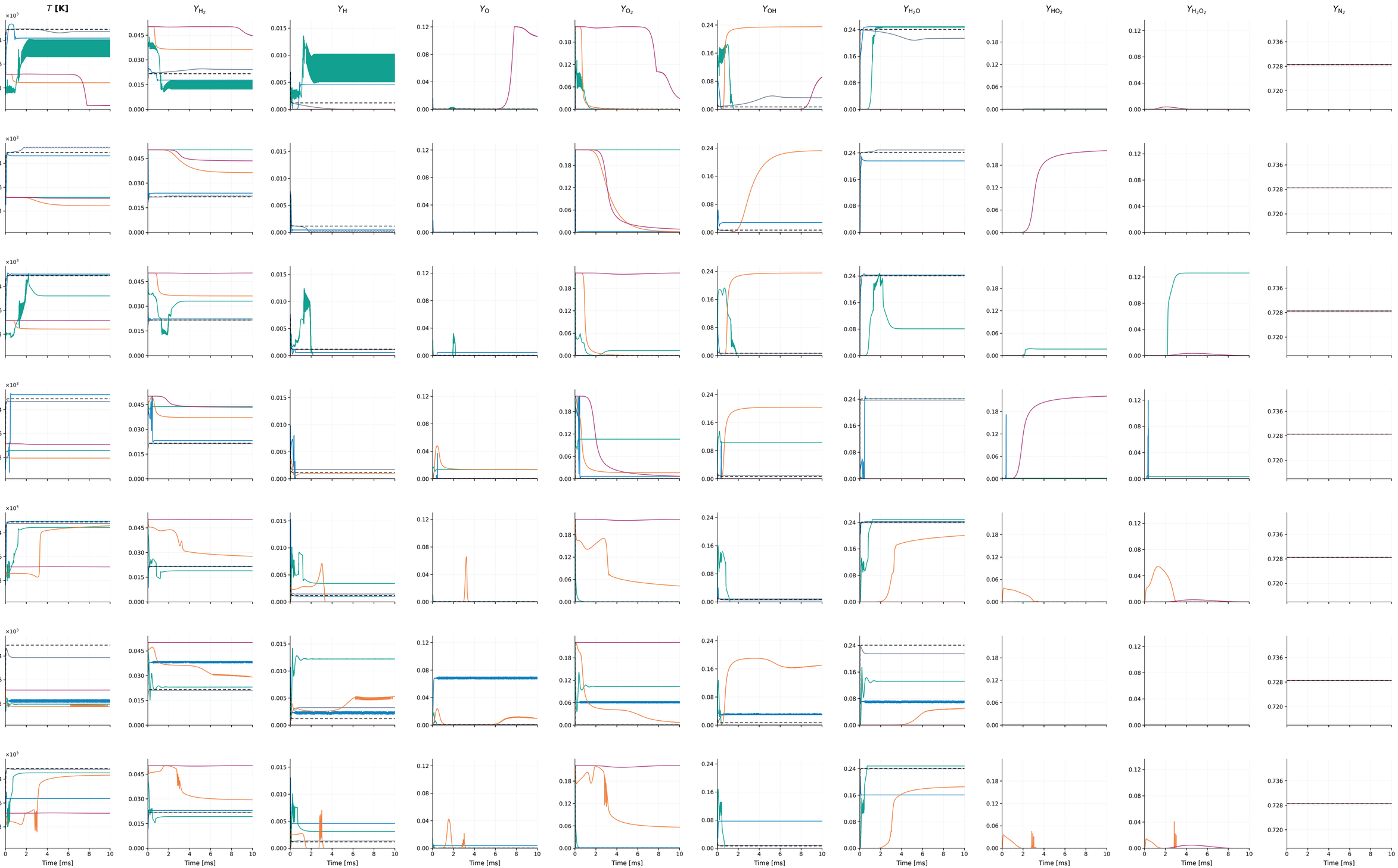
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 53 | Cantera vs NN NRMSE

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
1 ANN10	**1**	0.108	0.174	2.66	1.07	0.52	1.55	0.0561	6.74	665	0	Complete
1 ANN10	**10**	0.351	0.399	5.59	0.977	2.4	7.79	0.329	79.8	397	0	Complete
1 ANN10	**100**	0.64	0.74	1	1	5.93	31.9	1	1.18	182	0	Complete
1 ANN10	**1000**	0.66	1.29	1	113	20.4	3.29	1	0.999	6.95e+03	0	Complete
2 P-only DeepONet10	**0**	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	**1**	0.0545	0.126	0.661	1.43	1.14	3.18	0.118	0.997	3.15	0	Complete
2 P-only DeepONet10	**10**	0.545	1.32	1	1	23.3	1	1	0.908	0.929	0	Complete
2 P-only DeepONet10	**100**	0.613	0.933	1	1	12.3	25.4	1	1.18	3.84	0	Complete
2 P-only DeepONet10	**1000**	0.552	1.12	1	1	12.7	1	1	3.38e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0354	0.0549	0.522	8.6	0.598	1.2	0.0423	36.5	4.79	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.334	0.512	2.19	6.66	2.08	8.27	0.653	3.08e+03	5.41e+05	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.638	0.763	1	1	6.98	31.3	1	1.17	62.5	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.544	1.31	1	1	23.2	1	1	0.998	9.57e+03	0	Complete
4 T/P DeepONet10	**0**	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	**1**	0.146	0.208	1.21	2.27	4.02	1.13	0.184	1.1e+03	2.66e+04	0	Complete
4 T/P DeepONet10	**10**	0.628	1.02	0.993	26.2	11.1	14.1	1	369	1.70e+04	0	Complete
4 T/P DeepONet10	**100**	0.719	0.769	0.306	32.9	5.27	27.4	1	4.46	3.18	0	Complete
4 T/P DeepONet10	**1000**	0.553	1.08	1	1	9.98	1	1	3.70e+04	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.0383	0.0367	0.688	0.32	0.281	0.26	0.0152	0.549	0.884	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.185	0.164	2.43	0.491	1.03	5.08	0.175	51.1	427	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.366	0.63	1.39	15.4	10.4	1	0.571	2.87e+03	1.04e+05	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.544	1.31	1	1	23.2	1	1	0.998	8.78e+03	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	**1**	0.678	0.766	0.948	137	6.45	3.55	0.71	16.5	0.842	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.715	0.135	9.05	4.13	10.8	2.04	0.463	14.7	3.79	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.742	0.621	2.39	15.2	6.76	23.7	0.915	6.91	4.35	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.545	1.32	1	1	23.3	1	1	6.75	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.366	0.0672	2.85	7.61	0.209	10.3	0.332	1.09	1.91	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.139	0.145	1.99	0.798	0.952	3.63	0.135	68	375	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.352	0.746	1.09	15.1	12.8	1	0.587	1.8e+03	1.82e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.543	1.31	1	1	23.2	1	1	0.998	1.19e+04	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

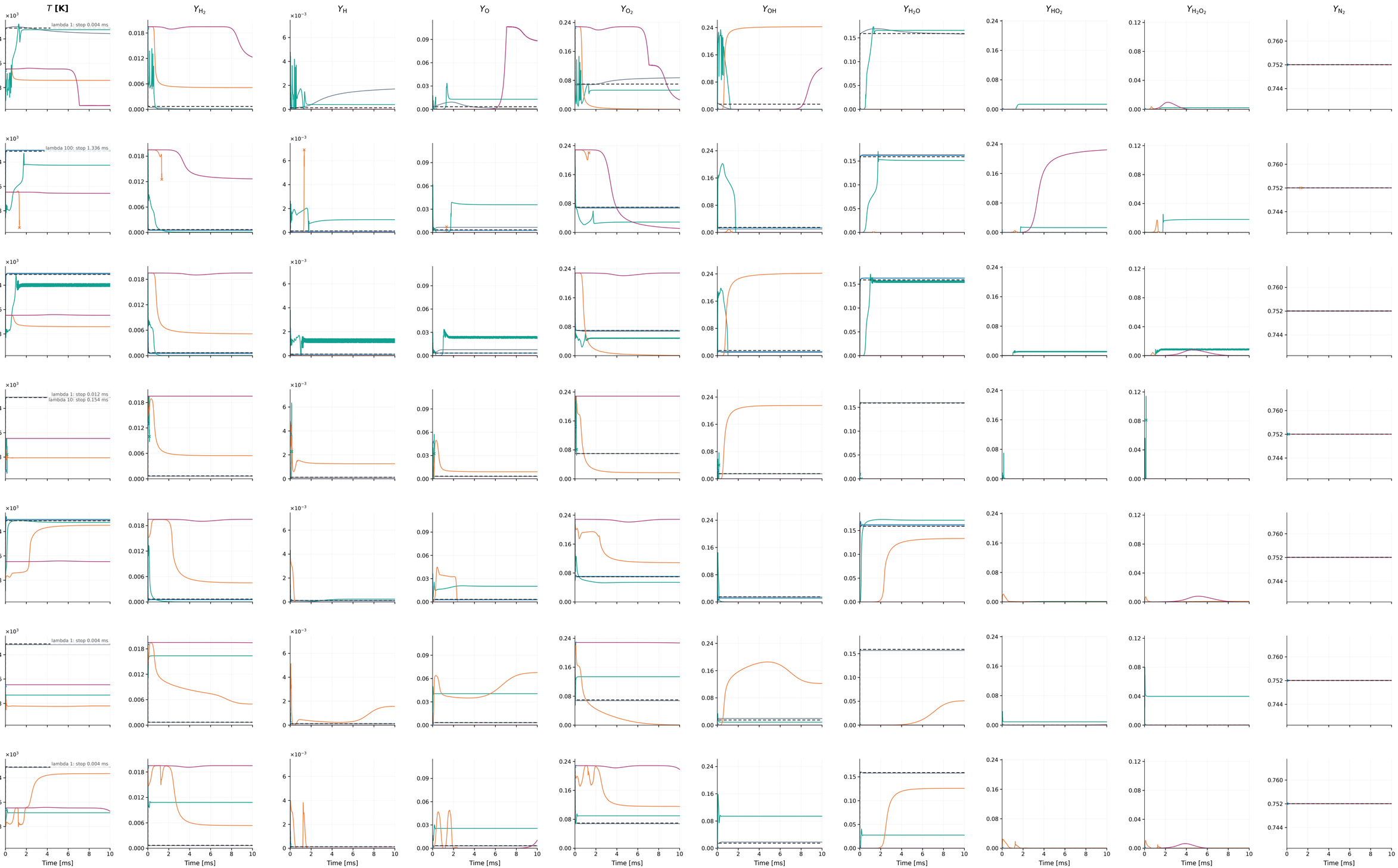
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 57 | Cantera vs NN NRMSE

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Repaired self-fed, 10 ms

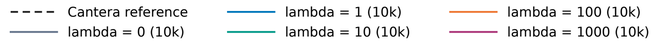
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

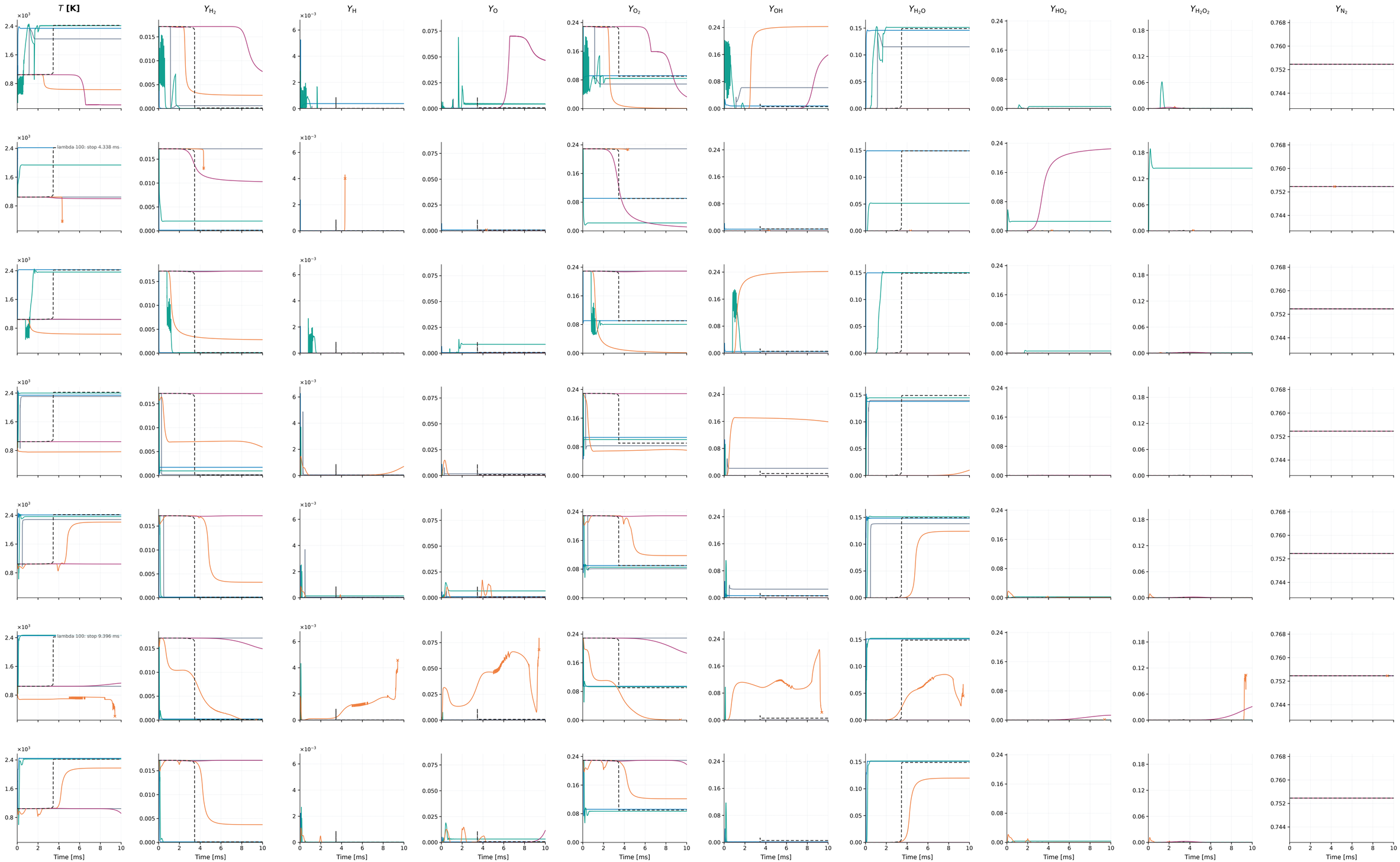
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
1 ANN10	**10**	0.178	2.17	4.29	2.86	0.305	3.41	0.261	470	1.22e+03	0	Complete
1 ANN10	**100**	0.613	7.97	1	1	1.08	14.4	1	1.09	269	0	Complete
1 ANN10	**1000**	0.655	21.7	1	15.9	1.87	2.3	1	0.985	1.93e+03	0	Complete
2 P-only DeepONet10	**0**	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	**1**	0.0213	0.623	0.279	0.214	0.0454	0.11	0.0337	0.279	0.303	0	Complete
2 P-only DeepONet10	**10**	0.266	2.27	8.16	8.9	0.588	4.05	0.307	449	9.7e+03	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 1.336 ms
2 P-only DeepONet10	**1000**	0.496	18	1	1	1.36	1	1	6.24e+03	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0252	0.645	0.208	0.181	0.0389	0.274	0.0414	0.201	0.443	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.211	2.11	8.65	5.82	0.331	3.19	0.252	376	4.71e+03	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.609	8.62	1	1	1.06	14	1	3.33	362	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.485	22.7	1	1	2.24	1	1	0.962	2.12e+03	0	Complete
4 T/P DeepONet10	**0**	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.154 ms
4 T/P DeepONet10	**100**	0.718	7.91	9.33	3.15	0.807	12.5	1	5.59	4.87	0	Complete
4 T/P DeepONet10	**1000**	0.487	22.8	1	1	2.27	1	1	0.561	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.024	0.426	0.429	0.289	0.0326	0.28	0.0326	0.696	0.18	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.0707	2.01	1.01	4.91	0.24	1.16	0.136	49	7.86	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.308	11.7	3.88	4.23	1.02	1	0.512	127	610	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.485	22.6	1	1	2.24	1	1	0.927	2.15e+03	0	Complete
6 T/P/ϕ DeepONet10	**0**	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	**10**	0.61	19	1.17	11.1	0.915	0.425	1	332	2.39e+04	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.742	10.2	6.01	13.8	0.844	9.14	0.907	2.5	2.49	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.487	22.8	0.999	1	2.27	1	1	12.1	24.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.546	12.3	1.25	6.65	0.295	5.28	0.827	3.29	2.89	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.307	12.1	5.05	3.52	1.15	1	0.537	163	789	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.488	22.7	1	0.973	2.24	1	1	0.93	1.44e+03	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 64 | Cantera vs NN NRMSE

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.284	0.791	1.02	1.04	0.509	11.4	0.523	1.02	1.01	0	Complete
1 ANN10	**1**	0.367	0.991	20.5	0.681	0.523	1.12	0.702	3.92	0.977	0	Complete
1 ANN10	**10**	0.323	0.9	15.3	8.44	0.551	8.46	0.608	86.6	57.5	0	Complete
1 ANN10	**100**	0.706	0.434	1	1	0.608	43.1	1	0.762	3.48	0	Complete
1 ANN10	**1000**	0.743	1.24	1	54.8	0.529	10.5	1	1	6.4	0	Complete
2 P-only DeepONet10	**0**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	**1**	0.391	0.987	2.16	0.93	0.527	0.659	0.722	1.02	0.999	0	Complete
2 P-only DeepONet10	**10**	0.312	0.876	1	1	0.865	1	0.697	451	970	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 4.338 ms
2 P-only DeepONet10	**1000**	0.558	0.865	1	1	0.361	1	1	2.9e+03	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.393	0.988	2.18	0.703	0.527	0.666	0.724	0.916	0.996	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.29	0.823	15	10.9	0.507	7.59	0.562	91.6	5.43	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.708	0.634	1	1	0.768	45.9	1	0.98	1.66	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.541	1.37	1	0.993	0.73	1	1	0.998	6.28	0	Complete
4 T/P DeepONet10	**0**	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	**1**	0.368	0.906	8.67	1.14	0.477	1.28	0.671	0.892	0.954	0	Complete
4 T/P DeepONet10	**10**	0.381	0.935	9.45	1.06	0.491	1.35	0.694	1.02	0.991	0	Complete
4 T/P DeepONet10	**100**	0.659	0.769	14.6	3.73	0.567	34.1	0.989	1	1	0	Complete
4 T/P DeepONet10	**1000**	0.542	1.37	1	1	0.73	1	1	9.66	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.331	0.919	4.92	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.387	0.985	3.42	0.899	0.529	0.933	0.717	0.898	0.998	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.365	0.974	11	9.5	0.541	2.07	0.712	46.6	1.97	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.258	0.638	6.14	4.8	0.333	1	0.462	54.8	7.83	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.541	1.37	1	0.995	0.73	1	1	0.997	5.39	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**1**	0.399	0.978	1.03	0.793	0.512	1	0.73	0.987	1	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.399	0.98	11.9	1.12	0.513	1.44	0.728	1.81	1.61	0	Complete
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 9.396 ms
6 T/P/ϕ DeepONet10	**1000**	0.533	1.32	0.994	1	0.668	1	1	101	67.9	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.395	0.983	2.89	0.768	0.519	0.909	0.726	0.894	0.998	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.379	0.973	7.91	5.32	0.535	2.14	0.715	60.2	2.96	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.208	0.536	7.08	5.16	0.29	1	0.383	60.5	8.64	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.545	1.37	1	2.51	0.727	1	1	0.996	6.26	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

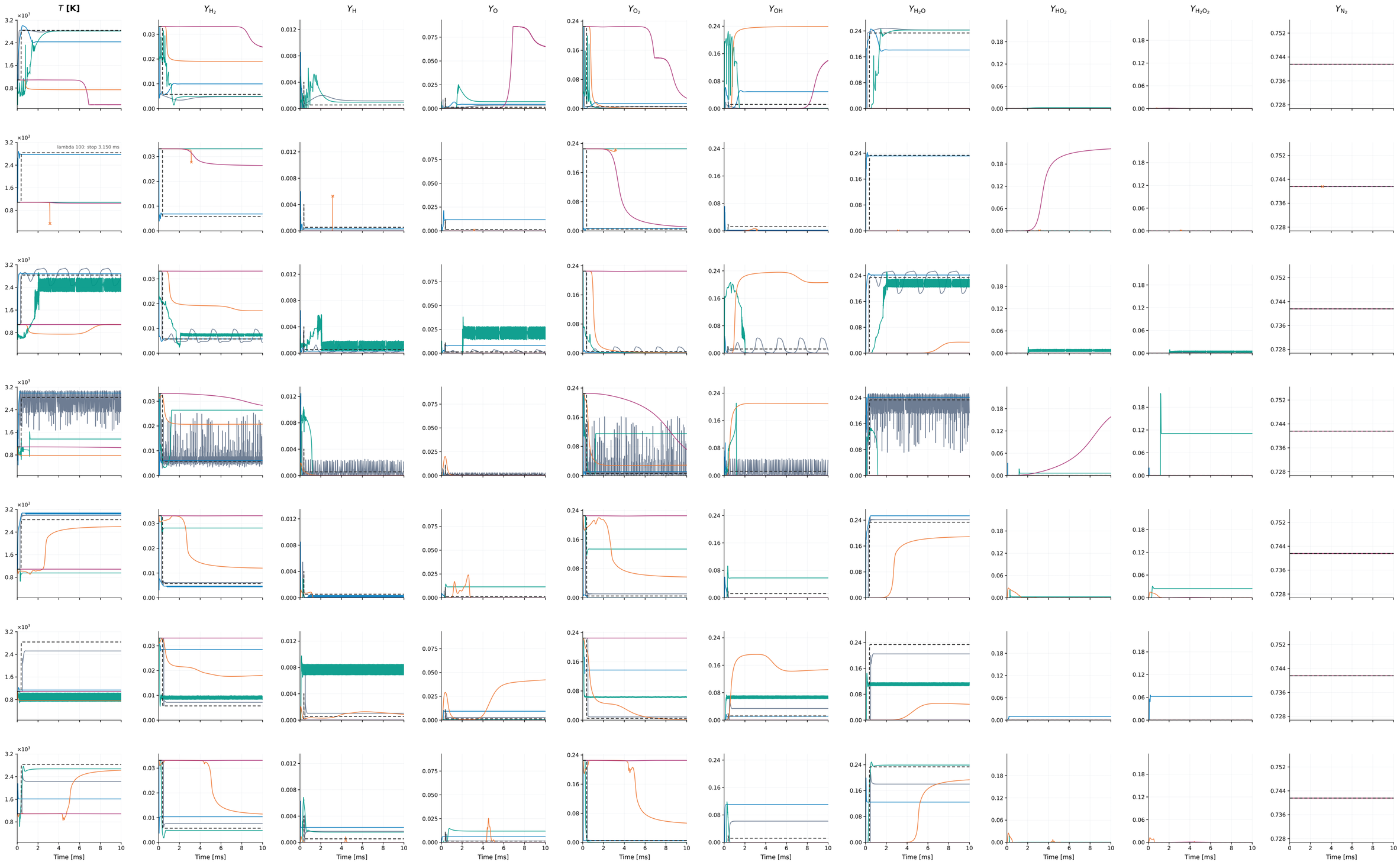
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 68 | Cantera vs NN NRMSE

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
1 ANN10	**1**	0.16	0.728	1.3	2.26	0.911	2.98	0.265	1.02	0.879	0	Complete
1 ANN10	**10**	0.21	0.484	2.4	5.36	0.82	4.91	0.242	91.8	16.8	0	Complete
1 ANN10	**100**	0.73	1.64	1	1	1	17.9	1	1.11	28.6	0	Complete
1 ANN10	**1000**	0.737	3.04	1	32.2	4.19	3.35	1	0.999	52.4	0	Complete
2 P-only DeepONet10	**0**	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	**1**	0.11	0.59	0.657	7.58	0.918	0.889	0.183	4.68	1.04	0	Complete
2 P-only DeepONet10	**10**	0.615	3.15	1	0.999	4.94	1	1	1.41	4.16	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 3.150 ms
2 P-only DeepONet10	**1000**	0.624	2.67	1	1	2.72	1	1	7e+03	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.0874	0.309	0.756	0.913	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.112	0.58	0.598	4.73	0.908	1.01	0.186	2.3	1.04	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.29	0.532	2.38	14.1	0.71	5.79	0.312	278	436	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.69	1.66	1	1	1.27	16.6	0.957	1.07	7.29	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.615	3.15	1	0.997	4.93	1	1	0.999	46.8	0	Complete
4 T/P DeepONet10	**0**	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	**1**	0.118	0.584	1.62	1.04	0.89	1.1	0.178	23.1	37.5	0	Complete
4 T/P DeepONet10	**10**	0.53	2.31	4.6	1	2.39	2.59	0.971	253	1.23e+04	0	Complete
4 T/P DeepONet10	**100**	0.722	1.77	1.06	2.5	0.793	15.8	1	1.03	1	0	Complete
4 T/P DeepONet10	**1000**	0.618	2.99	1	1	4.04	1	1	2.79e+03	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.129	0.587	0.974	1.06	0.913	1.09	0.193	1.01	1.33	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.662	2.58	1.01	7.32	2.89	3.78	1	118	2.82e+03	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.329	1.66	0.997	3.56	2.51	1	0.524	254	400	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.615	3.15	1	0.999	4.93	1	1	0.999	43.5	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	**1**	0.597	2.62	1	5.8	2.98	0.161	1	395	7.32e+03	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.688	0.614	13.2	0.704	1.44	4.55	0.531	1.21	1	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.73	1.64	0.96	20.4	0.919	12.2	0.881	1.02	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.615	3.15	1	1	4.94	1	1	0.964	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.432	0.728	3.24	3.41	0.909	8.19	0.478	1.22	0.993	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.0665	0.326	2.4	7.57	0.48	1.38	0.0599	34.2	3.36	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.438	2.27	1	2.46	3.53	1	0.714	152	238	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.615	3.15	1	0.999	4.93	1	1	0.998	61.1	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

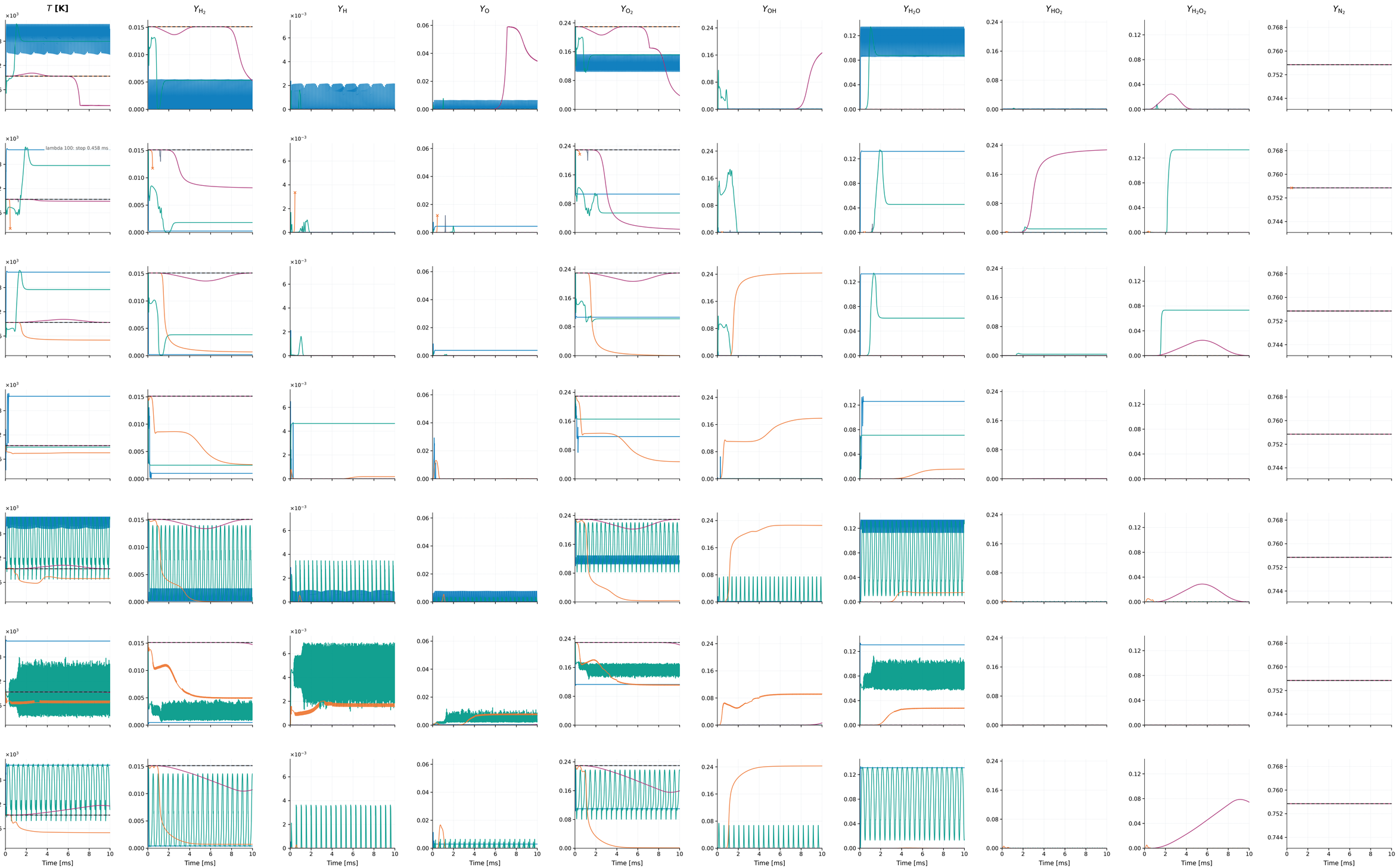
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 69 | Cantera vs NN NRMSE

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Repaired self-fed, 10 ms

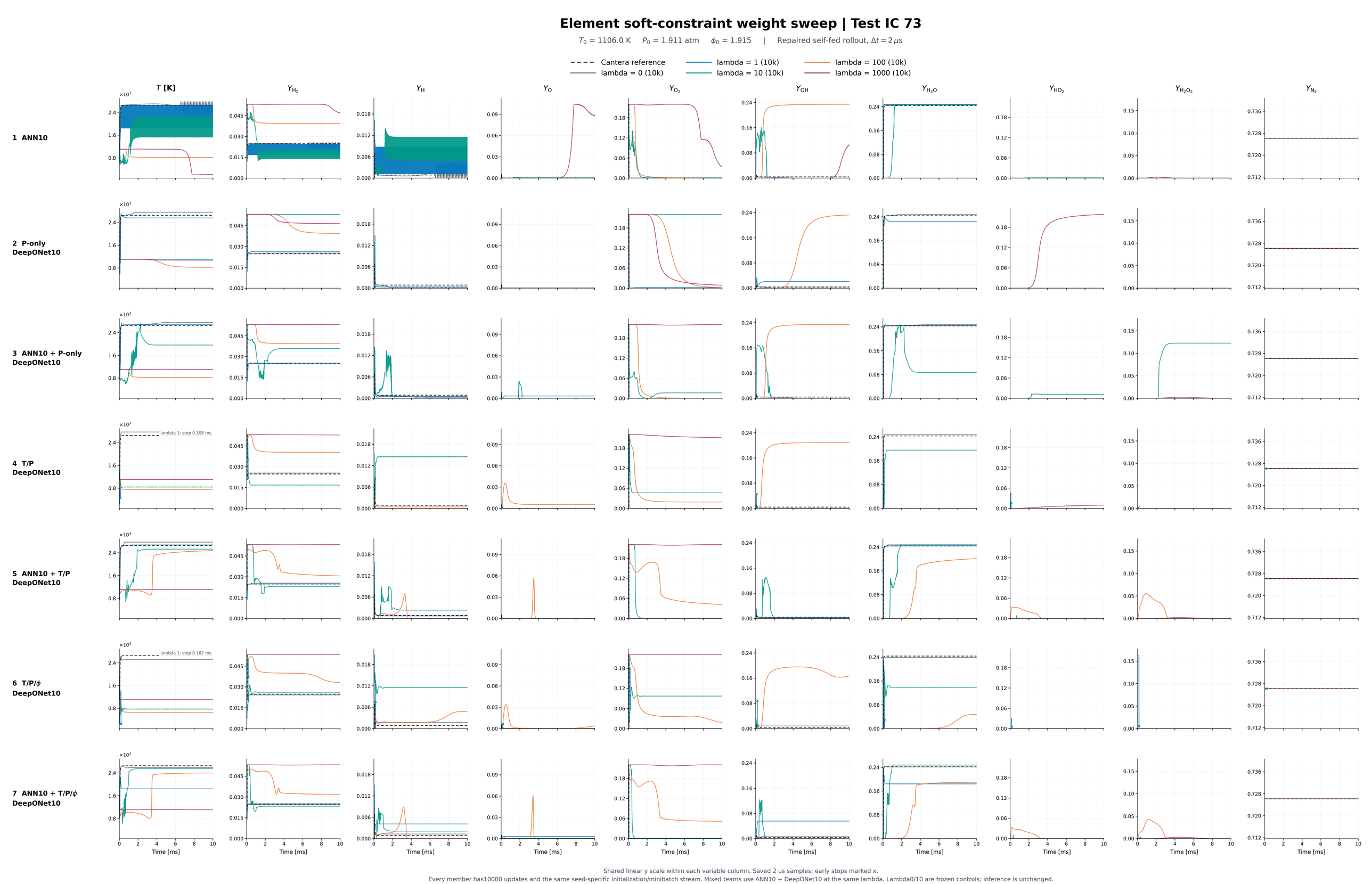
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
1 ANN10	**1**	1.17	0.89	1.79e+08	2.82e+08	0.468	2.36e+08	1.76e+06	2.52e+03	2.82e+03	0	Complete
1 ANN10	**10**	0.896	0.644	5.90e+07	4.44e+07	0.343	4.06e+09	1.28e+06	2.22e+03	8.1e+03	0	Complete
1 ANN10	**100**	9.75e-07	1.29e-06	9.98	87.1	1.56e-06	1	1	0.998	1	0	Complete
1 ANN10	**1000**	0.426	0.205	1	2.86e+09	0.3	1.45e+10	1	1.12	1.06e+05	0	Complete
2 P-only DeepONet10	**0**	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	**1**	1.31	0.98	1.37e+07	4.63e+08	0.534	3.09e+08	1.96e+06	1.15e+03	89.1	0	Complete
2 P-only DeepONet10	**10**	0.852	0.854	6.02e+07	4.02e+07	0.73	1.60e+10	7.59e+05	3.97e+04	1.33e+06	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.458 ms
2 P-only DeepONet10	**1000**	0.0405	0.361	1	1	0.756	1	1	7.95e+05	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	**1**	1.33	0.985	1.61e+07	3.94e+08	0.533	1.74e+08	1.98e+06	1.63e+03	114	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.85	0.735	7.06e+07	1.57e+07	0.538	7.93e+09	9.39e+05	1.69e+04	7.63e+05	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.413	0.846	227	67	0.885	6.06e+10	1	69.2	746	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.045	0.0536	1	987	0.0561	3.15e+06	1	1.23	1.56e+05	0	Complete
4 T/P DeepONet10	**0**	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	**1**	1.29	0.927	9.81e+07	2.14e+08	0.485	6.39e+08	1.87e+06	188	808	0	Complete
4 T/P DeepONet10	**10**	0.0397	0.832	1.48e+09	1.25e+05	0.276	1.91e+08	1.06e+06	59.3	1	0	Complete
4 T/P DeepONet10	**100**	0.194	0.628	4.32e+07	2.41e+08	0.616	3.99e+10	1.49e+05	1	1	0	Complete
4 T/P DeepONet10	**1000**	1.08e-04	9.71e-04	1	1	2.03e-03	1	1	2.14e+03	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1**	1.28	0.956	8.49e+07	3.04e+08	0.506	3.21e+08	1.91e+06	1.46e+03	720	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.872	0.714	3.22e+08	1.13e+08	0.393	6.85e+09	1.31e+06	1.85e+03	2.89e+03	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.266	0.878	2.05e+07	6.45e+07	0.874	5.69e+10	1.85e+05	2.78e+03	1.19e+04	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.0516	0.0616	1	1.23e+03	0.0644	2.90e+06	1	1.45	1.79e+05	0	Complete
6 T/P/ϕ DeepONet10	**0**	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	**1**	1.35	0.966	8.28e+05	3.95e+07	0.507	2.01e+06	1.95e+06	28.3	0.911	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.474	0.857	1.49e+09	5.38e+08	0.322	6.67e+06	1.17e+06	476	0.977	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.26	0.575	4.98e+08	6.22e+08	0.435	2.23e+10	3.39e+05	1	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	3.08e-03	4.29e-03	1	9.25e+06	4.87e-03	3.07e+08	1	103	1.29	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	9.40e-06	8.94e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	1.32	0.97	4.01e+06	3.20e+08	0.521	6.52e+07	1.95e+06	49	6.99	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.934	0.757	3.44e+08	2.12e+08	0.411	5.37e+09	1.41e+06	1.49e+03	1.26e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.422	0.865	8.71e+06	3.21e+08	0.904	6.19e+10	3.71e+03	3.22e+03	6.97e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.15	0.181	1	1	0.189	7.23e+05	1	3.62	5.25e+05	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.

Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 73 | Cantera vs NN NRMSE

T0 = 1106.0 K P0 = 1.911 atm phi0 = 1.915 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0381	0.0487	1.14	0.997	0.157	0.82	0.0187	0.373	6.14	0	Complete
1 ANN10	**1**	0.204	0.223	5.82	0.976	0.319	0.99	0.0311	9.53	221	0	Complete
1 ANN10	**10**	0.384	0.371	8.39	1.47	1.57	10.4	0.302	36.7	176	0	Complete
1 ANN10	**100**	0.682	0.636	1	1	3.2	54.2	1	1.03	59.3	0	Complete
1 ANN10	**1000**	0.684	1.11	1	199	10.9	6.27	1	1	1.85e+03	0	Complete
2 P-only DeepONet10	**0**	0.0376	0.025	0.757	0.734	0.154	0.77	0.019	0.37	0.648	0	Complete
2 P-only DeepONet10	**1**	0.0523	0.0791	1	1.2	0.344	4.14	0.0852	1.55	4.32	0	Complete
2 P-only DeepONet10	**10**	0.582	1.13	1	1	12.3	1	1	1	1	0	Complete
2 P-only DeepONet10	**100**	0.641	0.871	1	1	7.94	38.9	1	1	1.02	0	Complete
2 P-only DeepONet10	**1000**	0.59	0.958	1	1	6.65	1	1	1.69e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.03	0.0143	0.592	0.58	7.95e-03	0.633	0.0124	0.0794	0.188	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0288	0.0345	0.833	12.7	0.0845	0.954	6.57e-03	16.5	6.4	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.344	0.437	2.8	14.4	1.41	12.3	0.636	1.04e+03	2.27e+05	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.679	0.66	1	1	3.86	53.1	1	1.02	16.4	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.582	1.13	1	1	12.3	1	1	1	2.65e+03	0	Complete
4 T/P DeepONet10	**0**	0.0466	0.0258	0.956	0.772	0.0161	0.986	0.0171	0.459	6.48	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
4 T/P DeepONet10	**10**	0.681	0.331	14.7	0.991	2.85	0.99	0.218	8.18	0.969	0	Complete
4 T/P DeepONet10	**100**	0.714	0.666	0.53	37.1	2.86	48	1	1.05	1	0	Complete
4 T/P DeepONet10	**1000**	0.583	1.12	1	1	12	1	1	627	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0444	0.0276	0.93	0.673	0.0709	0.961	0.0159	0.422	2.4	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.0159	0.0735	0.765	0.663	0.541	0.331	0.0438	1.13	0.993	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.229	0.305	2.55	1	3.17	7.79	0.299	28.8	167	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.367	0.572	1.45	26.7	5.57	1	0.585	1.32e+03	4.76e+04	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.582	1.13	1	1	12.3	1	1	1	2.19e+03	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.0497	0.0333	0.96	1.13	0.242	1.05	0.0279	0.634	0.879	0	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.182 ms
6 T/P/ϕ DeepONet10	**10**	0.706	0.107	11.4	0.899	5.48	1.01	0.437	6.21	1.13	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.749	0.606	2.14	29.2	3.63	42.3	0.956	1.12	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.582	1.13	1	1	12.3	1	1	1.78	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0248	0.0245	0.472	0.561	0.21	0.58	0.0195	0.528	0.672	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.305	0.0749	3.57	12.3	0.57	12.8	0.247	1.11	1.13	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.159	0.204	2.05	0.985	2.09	4.76	0.196	41	414	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.382	0.563	2	29.7	5.77	1	0.573	1.2e+03	3.50e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.581	1.13	1	1	12.3	1	1	1	3.49e+03	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 74 | Cantera vs NN NRMSE

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Repaired self-fed, 10 ms

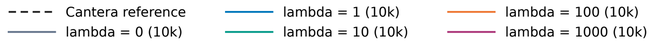
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

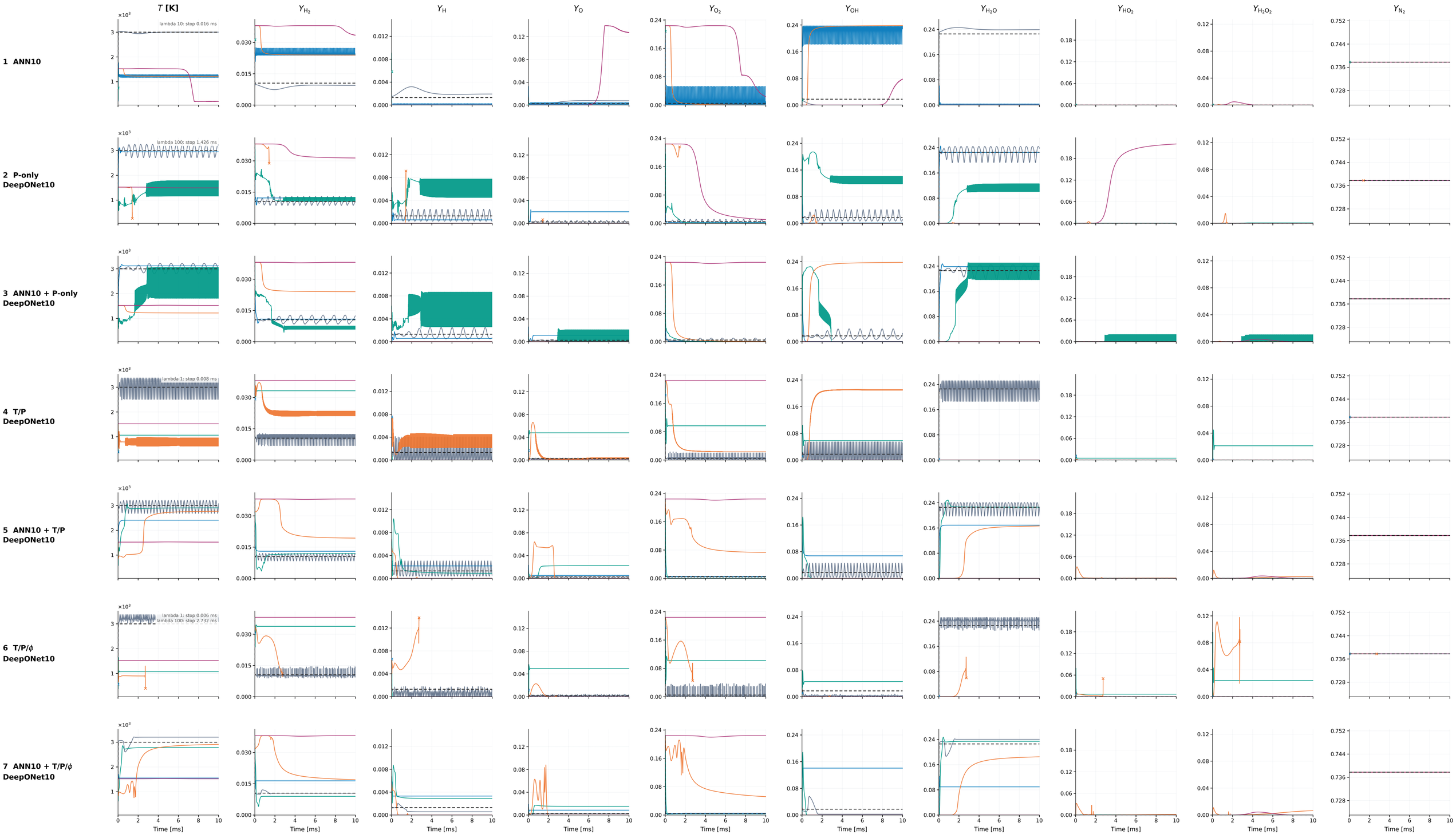
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
1 ANN10	**1**	0.0771	0.146	1.03	59.1	0.273	4.2	0.115	50.7	3.6e+03	0	Complete
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
1 ANN10	**100**	0.658	0.758	1	1	6.85	41.9	1	1.46	135	0	Complete
1 ANN10	**1000**	0.681	1.33	1	208	23.4	4.08	1	1	282	0	Complete
2 P-only DeepONet10	**0**	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	**1**	0.0425	0.0503	0.178	36.4	0.327	1.09	0.0324	3.05	3.37	0	Complete
2 P-only DeepONet10	**10**	0.469	0.272	7.4	3.92	2.78	14.3	0.386	13.6	734	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 1.848 ms
2 P-only DeepONet10	**1000**	0.57	1.16	1	1	15	1	1	1.93e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.022	0.0336	0.264	12.1	0.32	1.41	0.0334	1.24	6.16	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.447	0.277	6.44	1.57	2.51	15.6	0.411	1.89	78.6	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.599	0.744	1	1	7.85	37.2	0.927	1.36	33.7	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.563	1.35	1	1	26.8	1	1	0.999	333	0	Complete
4 T/P DeepONet10	**0**	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
4 T/P DeepONet10	**10**	0.488	1.06	1	3.99	14.1	1.21	1	1.53e+03	1.34e+05	0	Complete
4 T/P DeepONet10	**100**	0.731	0.791	1.37	25.5	8.37	32.1	1	6.45	2.11	0	Complete
4 T/P DeepONet10	**1000**	0.563	1.35	1	1	26.8	1	1	1.11	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.0574	0.0276	1.07	1.38	0.274	1.42	0.0359	0.966	1.63	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.676	1.09	0.163	99.1	13.9	7.54	1	1.41e+03	3.82e+04	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.37	0.752	1.07	74.1	12.9	1	0.564	914	4.64e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.563	1.35	1	0.999	26.8	1	1	0.999	136	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
6 T/P/ϕ DeepONet10	**10**	0.486	1.04	0.995	7	13.2	1.38	1	2.04e+03	1.39e+05	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.736	0.549	6.4	1.15	8.71	24.5	0.811	7.76	3.08	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.563	1.35	1	1	26.8	1	1	1.15	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.468	0.31	2.06	26.2	0.365	22.8	0.562	1.77	2.16	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.751	0.308	6.79	1.02	3.79	27	0.751	52.2	45.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.345	0.709	1.06	64.9	12.2	1	0.53	820	3.44e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.563	1.35	1	0.996	26.8	1	1	0.999	496	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 76 | Cantera vs NN NRMSE

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

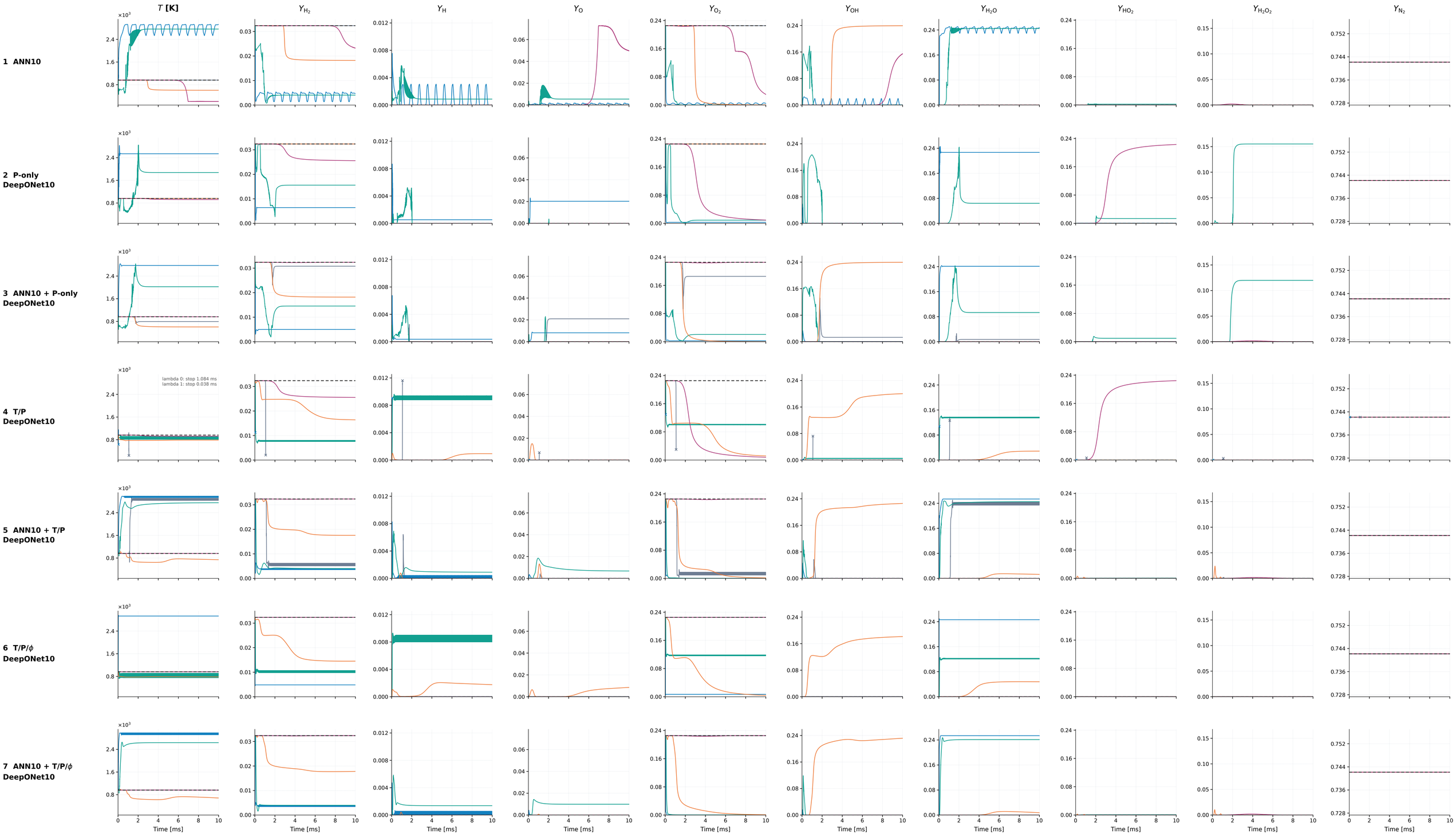
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
1 ANN10	**1**	0.589	1.35	0.965	0.944	2.61	11.8	0.995	2.84	0.885	0	Complete
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
1 ANN10	**100**	0.587	1.41	1	1	8.01	12	1	1.6	164	0	Complete
1 ANN10	**1000**	0.641	2.57	1	26.9	28.6	1.27	1	0.971	2.03e+03	0	Complete
2 P-only DeepONet10	**0**	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	**1**	0.0305	0.182	0.529	7.03	0.971	0.944	0.0413	24.9	1.56	0	Complete
2 P-only DeepONet10	**10**	0.567	0.463	3.76	1.08	2.36	7.31	0.609	19.4	427	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 1.426 ms
2 P-only DeepONet10	**1000**	0.498	2.24	1	1	19	1	1	2.53e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0521	0.112	0.553	3.43	0.925	1.09	0.0812	8.18	2.23	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.38	0.524	3.64	4.15	1.73	4.74	0.406	1.59e+03	6.13e+03	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.584	1.44	1	1	8.95	11.8	1	3.47	139	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.491	2.61	1	1	33.3	1	1	0.927	2.1e+03	0	Complete
4 T/P DeepONet10	**0**	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
4 T/P DeepONet10	**10**	0.646	2.16	0.995	18	14	2.33	1	783	2.67e+04	0	Complete
4 T/P DeepONet10	**100**	0.736	1.28	1.71	5.25	8.01	10.4	1	27.6	20	0	Complete
4 T/P DeepONet10	**1000**	0.492	2.62	1	1	33.5	1	1	1.56	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.2	0.251	0.669	1.11	0.773	2.9	0.256	5.12	2.84	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.12	0.253	1.38	7.42	1.72	1.42	0.116	44	52.4	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.342	1.49	1.08	9.6	16.2	1	0.561	829	2.88e+03	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.491	2.61	1	1	33.3	1	1	0.902	2.03e+03	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.006 ms
6 T/P/ϕ DeepONet10	**10**	0.643	2.2	1	18.7	14.9	1.61	1	1.13e+03	3.05e+04	0	Complete
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 2.732 ms
6 T/P/ϕ DeepONet10	**1000**	0.492	2.62	1	1	33.5	1	1	0.529	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.482	0.565	1.55	2.41	0.774	7.02	0.604	4.92	0.898	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.119	0.229	1.5	5.03	1.75	1.37	0.113	108	56.1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.287	1.32	1.07	6.63	15.2	1	0.497	827	4.32e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.491	2.61	1	1	33.3	1	1	0.899	2.4e+03	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.

Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 78 | Cantera vs NN NRMSE

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Repaired self-fed, 10 ms

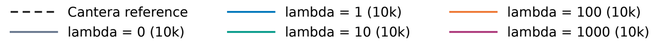
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

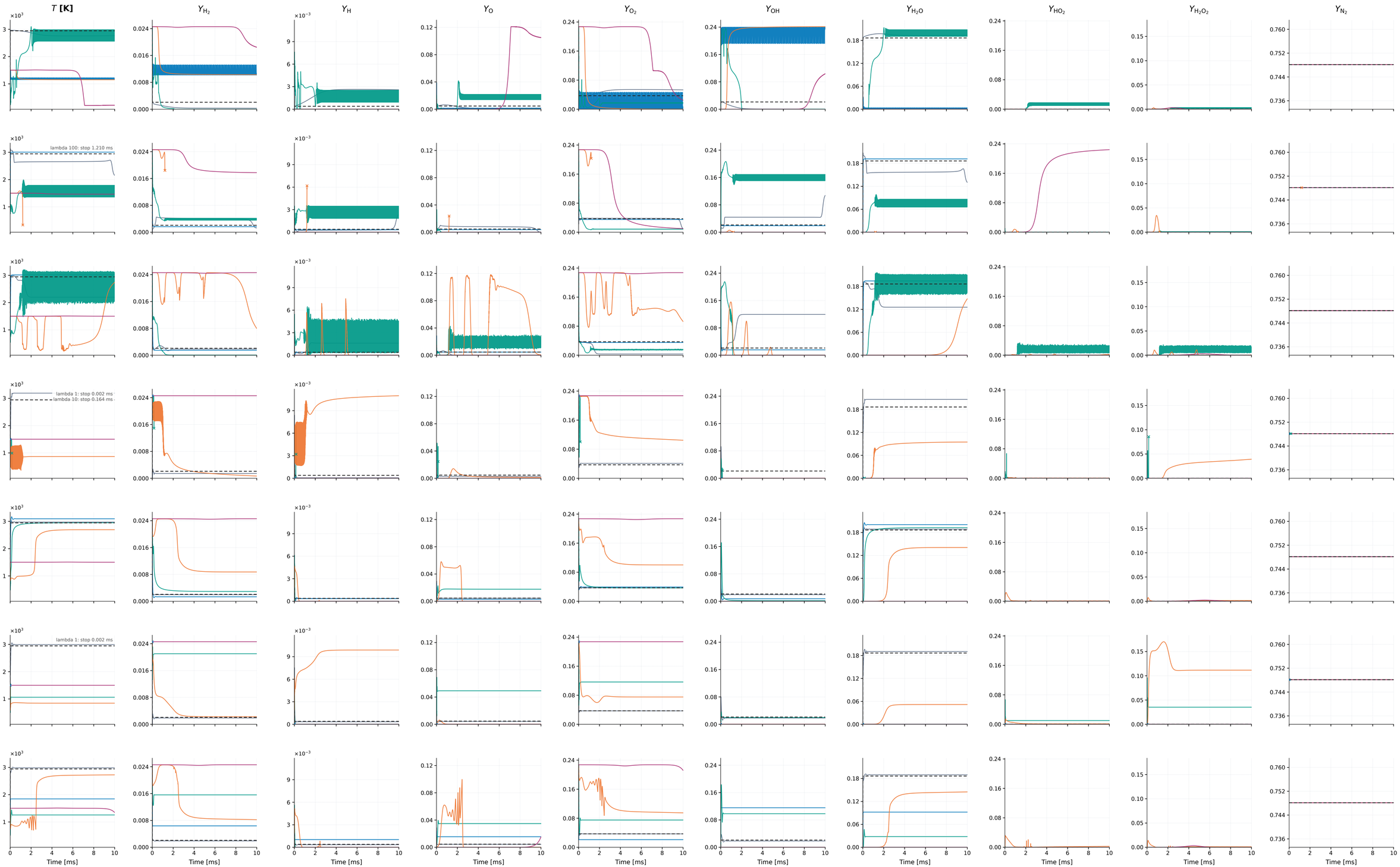
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
1 ANN10	**1**	1.9	0.873	7.43e+06	3.74e+06	0.981	6.05e+07	6.46e+04	16.7	6.24	0	Complete
1 ANN10	**10**	1.75	0.844	7.46e+06	2.11e+07	0.97	2.88e+08	6.18e+04	480	34.4	0	Complete
1 ANN10	**100**	0.307	0.361	1.01	1.03	0.824	1.37e+09	1	1.93	34.6	0	Complete
1 ANN10	**1000**	0.441	0.088	1	1.27e+08	0.324	3.33e+08	1	1	160	0	Complete
2 P-only DeepONet10	**0**	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	**1**	1.63	0.8	3.47e+06	7.04e+07	0.983	1.63e+07	6.03e+04	98.6	2.52	0	Complete
2 P-only DeepONet10	**10**	0.881	0.522	5.79e+06	6.43e+05	0.93	4.64e+08	1.83e+04	2.63e+03	2.96e+04	0	Complete
2 P-only DeepONet10	**100**	5.31e-05	2.47e-05	2.04	3.62e+03	4.84e-05	1	0.912	1	1	0	Complete
2 P-only DeepONet10	**1000**	0.0319	0.164	1	1	0.746	1	1	3.68e+04	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	**1**	1.88	0.843	2.52e+06	2.81e+07	0.985	1.85e+07	6.40e+04	46.1	1.02	0	Complete
3 ANN10 + P-only DeepONet10	**10**	1.06	0.561	5.41e+06	6.36e+06	0.894	3.85e+08	2.64e+04	1.89e+03	2.31e+04	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.331	0.39	1	1	0.889	1.48e+09	1	1.69	7.98	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	2.37e-03	1.60e-03	1	1.23e+04	3.65e-03	3.17e+03	1	1	188	0	Complete
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.084 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
4 T/P DeepONet10	**10**	0.112	0.756	4.75e+07	3.2e+03	0.552	3.23e+07	3.61e+04	2.57	1	0	Complete
4 T/P DeepONet10	**100**	0.186	0.366	2.94e+06	9.04e+06	0.741	1.13e+09	4.68e+03	1	1	0	Complete
4 T/P DeepONet10	**1000**	0.0339	0.174	1	1	0.793	1	1	3.92e+04	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	**1**	2.07	0.879	2.17e+06	1.28e+06	0.996	2.49e+07	6.73e+04	2.05	1.81	0	Complete
5 ANN10 + T/P DeepONet10	**10**	1.79	0.868	7.14e+06	2.99e+07	0.992	8.47e+07	6.41e+04	137	11.4	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.244	0.396	4.90e+05	6.27e+06	0.87	1.39e+09	2.62e+03	153	484	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	2.40e-03	1.62e-03	1	2.14e+03	3.69e-03	2.04e+03	1	1	191	0	Complete
6 T/P/ϕ DeepONet10	**0**	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**1**	2.04	0.85	1.02e+05	8.15e+04	0.969	4.23e+04	6.56e+04	26.1	8.75	0	Complete
6 T/P/ϕ DeepONet10	**10**	0.156	0.681	4.44e+07	8.67e+04	0.477	7.00e+04	3.23e+04	0.726	1	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.184	0.452	8.08e+06	1.75e+07	0.806	1.08e+09	9.86e+03	1	1	0	Complete
6 T/P/ϕ DeepONet10	**1000**	4.85e-05	2.03e-05	1	1	8.15e-05	1	1	4.69	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	2.07	0.881	1.66e+06	7.52e+05	0.997	6.82e+06	6.75e+04	1.54	2.3	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	1.71	0.872	8.09e+06	3.49e+07	0.991	7.41e+07	6.36e+04	186	7.66	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.277	0.404	2.30e+05	3.06e+05	0.898	1.45e+09	1.87e+03	64.8	203	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	2.54e-03	1.71e-03	1	2.06e+04	3.90e-03	2.45e+03	1	1	201	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 88 | Cantera vs NN NRMSE

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.0536	0.764	5.02	0.609	0.392	0.914	0.0409	1.27	4.48	0	Complete
1 ANN10	**1**	0.6	4.23	1.01	0.953	0.759	9.78	0.995	1.12	0.969	0	Complete
1 ANN10	**10**	0.224	1.3	4.59	2.8	0.599	2.9	0.282	449	1.05e+03	0	Complete
1 ANN10	**100**	0.6	4.64	1	1	1.52	9.93	1	1.06	196	0	Complete
1 ANN10	**1000**	0.658	10.3	1	13.1	4.24	1.42	1	0.981	509	0	Complete
2 P-only DeepONet10	**0**	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	**1**	0.0281	0.332	0.17	0.211	0.115	0.124	0.0419	0.529	0.621	0	Complete
2 P-only DeepONet10	**10**	0.489	1.35	6.09	1	0.775	6.68	0.625	6.29	356	0	Complete
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 1.210 ms
2 P-only DeepONet10	**1000**	0.498	8.64	1	1	2.81	1	1	6.21e+03	1	0	Complete
3 ANN10 + P-only DeepONet10	**0**	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	**1**	0.0352	0.411	0.0885	0.121	0.105	0.353	0.0607	0.413	0.797	0	Complete
3 ANN10 + P-only DeepONet10	**10**	0.249	1.3	6.19	3.35	0.612	2.55	0.27	558	5.32e+03	0	Complete
3 ANN10 + P-only DeepONet10	**100**	0.674	9.55	2.59	14	3.35	1.55	0.936	27.4	1.04e+03	0	Complete
3 ANN10 + P-only DeepONet10	**1000**	0.489	10.6	1	1	5.07	1	1	0.953	625	0	Complete
4 T/P DeepONet10	**0**	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.164 ms
4 T/P DeepONet10	**100**	0.709	2.81	24.8	0.764	2.57	1	0.58	19.8	1.41e+04	0	Complete
4 T/P DeepONet10	**1000**	0.49	10.6	1	1	5.09	1	1	0.908	1	0	Complete
5 ANN10 + T/P DeepONet10	**0**	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	**1**	0.0572	0.431	0.175	0.413	0.102	0.683	0.0843	0.592	0.501	0	Complete
5 ANN10 + T/P DeepONet10	**10**	0.0755	1.02	1.11	2.8	0.307	1.05	0.122	3.37	7.39	0	Complete
5 ANN10 + T/P DeepONet10	**100**	0.334	5.76	1.79	4.57	2.4	1	0.536	134	683	0	Complete
5 ANN10 + T/P DeepONet10	**1000**	0.489	10.6	1	1	5.07	1	1	0.879	496	0	Complete
6 T/P/ϕ DeepONet10	**0**	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
6 T/P/ϕ DeepONet10	**10**	0.639	8.9	1.05	9.61	2.12	0.0811	1	376	1.67e+04	0	Complete
6 T/P/ϕ DeepONet10	**100**	0.712	1.49	22.9	0.968	1.14	1	0.788	93.8	5.73e+04	0	Complete
6 T/P/ϕ DeepONet10	**1000**	0.49	10.6	1	1	5.09	1	1	0.833	0.997	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.374	2.04	1.65	2.35	0.435	4.47	0.507	0.627	0.0404	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.573	6.37	1.08	6.49	1.03	3.68	0.85	2.5	1.25	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.346	5.66	2.16	5.02	2.32	1	0.535	203	973	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.491	10.6	1	0.981	5.06	1	1	0.86	554	0	Complete

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.